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Entertainment games for complex problem-solving

**Implementing game design features for fostering
knowledge and skill acquisition**

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Summary

This thesis aims to design, develop, and evaluate a serious game intended for the training of Complex Problem Solving skills. Since formal learning environments struggle to simulate real-world complexity, this research explores the educational potential of the video game medium by applying the CPS-GFC methodological framework (Gyaurov, Fabricatore, and Bottino, 2025). The developed digital artifact consists of a text-based narrative adventure in which the player takes on the role of an isolated soldier in hostile territory. The game design employs specific Information Management Strategies (IMS) to modulate uncertainty, concealing or revealing crucial data to trigger specific cognitive processes. Within this inherently ill-defined ecosystem, the user is required to formulate hypotheses, manage risk exposure, and face ethical dilemmas, continuously balancing their moral compass against the rational optimization of resources. From a technical perspective, the prototype was developed in Unity, integrating the Ink scripting language to handle complex narrative branching alongside silent telemetric tracking. To evaluate the educational effectiveness of the experience, an exploratory study was conducted on a sample of 27 participants. The methodological framework cross-referenced objective data from the in-game telemetry with subjective perceptions measured via pre-test and post-test questionnaires. The results of the qualitative and quantitative analysis highlight a high degree of engagement and suggest, with statistical significance, the activation of prospective learning dynamics. When exposed to uncertainty, players indeed demonstrated the ability to internalize the system's opaque rules, effectively transferring acquired mental models to overcome novel challenges. Furthermore, the immersive power of the narrative appears to have fostered a solid internalization of agency, compelling users to take full responsibility for the irreversible consequences dictated by their choices. In conclusion, this study gathers encouraging empirical evidence supporting the effectiveness of the CPS-GFC framework. The data suggests that the strategic use of uncertainty, typical of entertainment titles, represents a promising path for architecting serious games capable of training complex cognitive skills, without having to sacrifice ludic engagement in favor of explicit didactics.

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*We played hide and seek in waterfalls
We were younger, we were younger
Someday we will foresee obstacles
Through the blizzard, through the blizzard*

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Acronyms

API

Application Programming Interface

CPS-GFC

Complex Problem Solving - Game Feature Configurations

HUD

Heads-Up Display

IMS

Information Management Strategies

IQR

Interquartile Range

NPC

Non-Player Character

PE

Player Engagement

RQ

Research Question

UI

User Interface

UX

User Experience

Chapter 1

Introduction

1.1 Context: Complex Problem Solving and the Educational Challenge

In the contemporary landscape, the ability to address uncertain and dynamic situations has emerged as an indispensable transversal skill. The scientific literature identifies this ability as Complex Problem Solving (CPS), rigorously defining it as “a collection of self-regulated psychological processes and activities necessary in dynamic environments to achieve ill-defined goals that cannot be reached by routine actions” [1]. As argued by Herbert A. Simon in his work *Models of Man: Social and Rational* (1957), cited by Mainzer (2009) [2], CPS represents one of the most critical skills required for the future, since “the capacity of the human mind for formulating and solving complex problems is very small compared with the size of the problem whose solution is required”.

Unlike “well-defined” problems, which have “a clear set of means for reaching a precisely described goal state” [1], complex problems are intrinsically “ill-defined”. Dörner and Funke [1], systematizing previous studies in the field, outline five typical attributes that distinguish a complex system:

- **Complexity:** the “sheer number of involved variables”;
- **Connectivity:** variables exhibit “mutual dependencies”, whereby a change in one part of the system affects the others;
- **Dynamics:** the situation reflects “the role of time and developments within a system”, which can change even without the solver’s intervention;
- **Intransparency:** there is a lack of clarity, either partial or total, “about the involved variables and their current values”;

- **Polytely:** the presence of “many goals”, which often represent conflicts at different levels of analysis.

These characteristics require the solver not only to apply prior knowledge but also to acquire new information through direct interaction with the system. The dynamic and intransparent nature of such problems makes it difficult to develop “cognitive CPS capabilities in formal learning environments”, as highlighted by Gyaurov, Fabricatore, and Bottino (2025) [3]. It is within this context that digital technologies, particularly simulated systems and games, emerge as promising tools for CPS research and training: indeed, they can create complex interactive scenarios capable of exposing the user to interconnected variables and unforeseen reactions within a controlled environment.

1.2 Research Problem

While digital environments offer the ideal space to simulate complexity, the core of the cognitive challenge for the user lies in managing uncertainty. To solve a complex problem, individuals cannot rely on predefined algorithms or mere routine; instead, they must engage in a continuous process of acquiring and applying knowledge towards a specific goal (knowledge acquisition and goal-oriented knowledge application) [4]. This process requires bridging the gap between the intransparent current state and the desired goal through the active acquisition of information, the simplification through reduction to the essential, the construction of a model of connectivity to handle the involved variables, the activation of dynamic decision making, and an evaluation phase [4].

For the individual to be driven to autonomously trigger and sustain these cognitive processes, it is essential that the interactive environment provides profound meaning to their actions, going beyond mere exposure to a challenging problem. Indeed, the literature suggests that the effectiveness of entertainment games in promoting CPS lies in their ability to immerse the player in cycles of *meaning-making*. As highlighted by Fabricatore (2018) [5], “if we accept that games can be regarded as systems, then we should also regard players as system thinkers who play through making sense of things, consciously and subconsciously interpreting meanings and establishing relationships to understand what has happened in the past, what is happening in the present, and predict what could happen in the future”. This systemic understanding of mechanics and context is not only a functional requirement but also constitutes the primary source of the user’s intrinsic motivation [5].

The need to trigger and support this motivation leads to the core of the research problem: the adoption of games specifically designed for education has often failed to promote CPS due to a design inability to balance player engagement with the

game’s high cognitive demand [3, 6]. Conversely, entertainment games excel at engaging players in highly motivating experiences that foster gameplay learning, a form of learning “intrinsically motivated by the game goals, contextualizations and mechanics” [3, 7, 8]. Gyaurov, Fabricatore, and Bottino (2025) [3] suggest that a potential solution to bridge the gap between educational and entertainment games lies in leveraging the largely unexplored potential of applying specific design guidelines – originally conceived for or derived from entertainment titles – to serious games.

Solving this research problem thus requires overcoming a design hurdle: the difficulty of translating cognitive theories into interactive gaming experiences by operationalizing the complex psychological and decision-making processes associated with CPS within concrete game mechanics [3]. The guiding question of this work is therefore how gameplay features should be structured in order to intentionally modulate uncertainty, compelling the player to manage incomplete information within an intrinsically motivating environment.

1.3 Thesis Objectives

To address the outlined research question, the present work is configured as an exploratory study whose main objective is the design and development of a serious game specifically aimed at training CPS skills. Specifically, the developed artifact is designed as a text-based narrative adventure featuring point-and-click mechanics. This game genre was chosen due to its natural propensity to confront the solver with complex decision trees governed by systemic relationships and long-term consequences, compelling them to balance conflicting priorities and objectives (polytely).

The project aims to overcome the limitations found in traditional learning environments by creating an interactive space where the player is constantly exposed to intrinsically ill-defined situations characterized by incomplete information. The foundational intent of this thesis, however, does not end with the creation of a difficult-to-solve ludic obstacle. On the contrary, the central objective is to implement specific game design elements tailored to intentionally modulate uncertainty and the availability of information within the game environment.

Through this strategic manipulation, the serious game aims to structure an experience that compels the user not to passively undergo events, but rather to proactively manage environmental and narrative uncertainty. The ultimate goal is to ensure that resolving the game’s challenges requires the sequential triggering of the cognitive processes typical of uncertainty management in CPS: the perception of an intransparent environment, the constant formulation of hypotheses to bridge information gaps, the selection of the most appropriate strategy, and, finally, the

field-testing of one’s deductions. In this way, the work intends to explore how uncertainty can cease to be a factor generating mere frustration, evaluating whether and to what extent it can transform into the primary engine of gameplay learning and player engagement.

1.4 Methodology: The CPS-GFC Framework

To operationalize the outlined objective and translate the need to modulate uncertainty into concrete game mechanics, the artifact’s design is rigorously based on the methodological framework proposed by Gyaurov, Fabricatore, and Bottino (2025) [3]. This study provides a theoretical and practical construct aimed at identifying how specific features of entertainment games can simultaneously promote player engagement and the development of CPS-related cognitive skills. The design foundation of this thesis relies, in particular, on the application of the ten guidelines termed CPS-GFC (*Complex Problem Solving - Game Feature Configurations*), outlined by the authors to support the development and evaluation of complex game environments.

The beating heart of these guidelines lies in the systematic implementation of the so-called *Information Management Strategies* (IMS) [3]. These strategies provide the game designer with a formal model that focuses on “increasing or decreasing player uncertainty by revealing or concealing information concerning the four key aspects of an uncertain phenomenon, i.e., presence, causes, effects and/or rules” [3].

Specifically, IMS define two complementary approaches. When the design strategically conceals information (thereby increasing uncertainty), the player is actively called upon to compensate for this cognitive gap. This compels the user to trigger the fundamental sub-processes of uncertainty management in CPS: problem perception, hypothesis formulation, selection of the best strategy, and the field-testing of their deductions. Conversely, when the game strategically reveals information, it provides the necessary support for the player to validate their intuitions. By intentionally alternating these phases of concealment and revelation, the CPS-GFC guidelines make it possible to balance the required cognitive load with adequate gratification, keeping user engagement intact.

The present thesis establishes this framework as the methodological foundation of the entire development process. The ten CPS-GFC operational configurations serve as the architecture for structuring the adventure’s decision points and complex situations. In this way, every interaction, dialogue, or resource management mechanic within the narrative environment is rigorously mapped onto specific information management strategies. This methodological rigor ensures that the entire architecture of the gaming experience is aimed at and validated against the

goal of training the player to solve complex problems.

1.5 Case Study

The described methodological architecture finds its practical application in the development of an interactive prototype, provisionally titled *Oltre Il Confine* (Beyond the Line). The artifact immerses the player in a hostile and morally gray setting: the protagonist is a soldier who, having survived a crushing defeat, finds himself stranded and isolated behind enemy lines. The long-term objective of the journey is the return home, a narrative device that serves as a framework for a series of interconnected systemic challenges.

From a thematic and gameplay perspective, the core of the experience revolves around the management of scarce resources (such as food, medical equipment, and tools), survival in a lethal environment, and the need to make moral choices under conditions of profound uncertainty. The central theme of the narrative is the deconstruction of the psychological distance between the protagonist and “the enemy”, between “us” and “them”. This thematic exploration is primarily conducted through interaction with two key figures: Elias, a deserter from the enemy army, and Lira, a spy belonging to the protagonist’s faction, yet opportunistic and ambiguous. However, this deconstruction is not limited solely to the presence of these companions: even if the player chooses a path of total isolation, forgoing their support, the game continues to challenge the concept of the “enemy” through the long-term consequences of the moral choices made and the exploration of specific narrative sections. In these instances, the design demonstrates how the brutality of war and shared suffering can blur the ideological boundaries between factions.

These narrative elements do not simply serve as a backdrop, but rather constitute the practical implementation of CPS within the game. The involved actors, with their often conflicting interests, generate a condition of polytely: the solver is constantly forced to balance their own survival, the safety of the potential group, and their personal moral compass. In this scenario, social and environmental interactions directly implement the CPS-GFC guidelines. For example, Elias, Lira, and many of the NPCs encountered embody the *Unpredictable Characters* guideline: the design strategically conceals the real causes or long-term effects of their behaviors, compelling the player to formulate and test hypotheses about their true intentions. Similarly, environmental exploration is governed by the *Risk-Reward* guideline, requiring the player to evaluate whether the retrieval of a vital resource is worth the exposure to unexpected events (e.g., encountering enemy patrols or crossing unstable ruins).

From a technical standpoint, the development of the serious game utilizes the Unity game engine, integrated with the Ink narrative scripting language. This

software toolset is particularly well-suited to two fundamental purposes of the thesis. First, Ink is specifically designed to agilely handle complex branching narrative structures, an essential feature for simulating the decision trees and cascading consequences typical of complex problems. Second, the hybrid architecture based on C# allows for granular and silent tracking of gameplay metrics at runtime. This telemetry infrastructure provides a fundamental objective baseline for the evaluation and analysis phase of the data collected during testing.

Thanks to this technical-narrative synergy, the entire adventure has been divided and organized into a series of six sequential ludonarrative scenarios, which are in turn subdivided into various branching sub-situations (nodes) whose interconnections depend on the player's choices. Each scenario represents an interactive ecosystem in which specific IMSs intentionally manipulate the available information. The choices made within a node organically affect subsequent ones according to the *Persistent Game World* guideline, ensuring that the player tangibly experiences the weight of their decisions within a dynamic and reactive system.

1.6 Validation Strategy

To investigate the potential of the proposed game architecture, this thesis work incorporates an exploratory evaluation phase based on a mixed-methods approach, which combines traditional playtesting sessions with the silent collection of objective metrics tracked at runtime by the system. In line with the research objectives, the evaluation strategy does not focus solely on the player's success or failure in overcoming the proposed challenges, but aims to investigate two fundamental and complementary dimensions of the user experience.

The first dimension concerns the player's level of engagement, which we have already discussed as an essential condition for triggering gameplay learning (Section 1.2). The validation aims to ascertain whether constant exposure to conditions of ambiguity generates a frustrating cognitive overload or whether, conversely, the application of the CPS-GFC guidelines succeeds in maintaining high motivation, transforming complexity into a stimulating challenge.

The second dimension investigates the perception of cognitive affordances, namely the ability of IMSs to offer users clear opportunities and stimuli to activate their mental processes. Specifically, the evaluation is designed to verify whether the game's design effectively conveyed to the participants the need to implement specific CPS strategies. The objective is to measure the players' propensity to engage in the sub-processes of uncertainty management, ascertaining whether the strategic concealment or revelation of information proactively pushed them towards the phases of problem perception, hypothesis formulation, strategy selection, and field testing.

To rigorously investigate these two dimensions, the analytical framework is structured to operate on two synergistic levels: the reconstruction of the global user experience and the comparative evaluation of individual ludonarrative situations. Combining subjective playtesting feedback (useful for measuring engagement and perceived cognitive effort) with the aforementioned objective behavioral metrics (narrative choices, paths taken, and resource management) enables the isolation and comparison of the effectiveness of different design choices. This convergence of qualitative and quantitative data constitutes the basis for gathering preliminary evidence and evaluating the extent to which the different configurations of uncertainty, applied in the various nodes of the adventure, support the training of CPS-related skills.

1.7 Thesis Outline

This thesis is structured along a logical path that proceeds from the theoretical review to the practical development of the project, concluding with an empirical analysis of the results. The work is organized as follows:

- **Chapter 2 – State of the Art:** defines the theoretical framework of reference, delving into the nature of Complex Problem Solving and the dynamics of meaning-making. It proposes an analysis of commercial titles to highlight the use of CPS in entertainment games and explores the methodological framework by Gyaurov, Fabricatore, and Bottino (2025) [3], defining the *Information Management Strategies* (IMS) and the CPS-GFC guidelines.
- **Chapter 3 – Design and Implementation:** illustrates the game design and the technical implementation of the prototype. It analyzes the translation of theory into game mechanics and describes the software architecture used to structure the narrative branches and modulate uncertainty.
- **Chapter 4 – Experimentation and Results:** presents the empirical validation methodology. It outlines the playtesting protocol, the combined use of runtime metrics and subjective feedback, and discusses the qualitative and quantitative analysis of the collected data.
- **Chapter 5 – Conclusion and Future Developments:** assesses the project against its initial objectives, critically analyzes the emerged limitations, and suggests possible directions for future research on the use of video games as cognitive training environments.

Chapter 2

State of the Art

This chapter outlines the theoretical and methodological foundations upon which the design of the artifact is built. Rather than proposing another systematic literature review regarding the educational potential of entertainment video games – a task already rigorously accomplished by the study of Gyaurov, Fabricatore, and Bottino (2025) [3] – this work adopts that research as its methodological anchor, avoiding its replication.

The discussion, therefore, focuses on the psychological and game design constructs strictly necessary for understanding the project. The following sections will first analyze the cognitive mechanisms underlying CPS and the motivational dynamics of gameplay learning. Subsequently, the CPS-GFC methodological framework [3] will be illustrated in detail, defining the guidelines and Information Management Strategies (IMS) that constitute the primary matrix of this thesis. Finally, the theoretical framework thus delineated will be employed as an analytical tool to examine a selection of commercial titles, demonstrating how the proposed operational configurations already find effective application within the entertainment industry.

2.1 Complex Problem Solving

As introduced in Chapter 1, the nature of Complex Problem Solving (CPS) diverges radically from traditional problem solving. The transition from well-defined scenarios to intrinsically ill-defined and intransparent environments does not entail a mere increase in mathematical or logical difficulty, but dictates a profound cognitive paradigm shift. As argued by Dörner and Funke (2017) [1], tackling a complex system requires moving beyond mere reasoning (logical-deductive reasoning based on certain evidence) to embrace thinking (divergent thinking). In situations where only a fraction of the necessary information is available, the solver must develop a “sense of possibility”, namely the ability to mentally formulate an unknown

reality, filling informational gaps through intelligent deductions without falling into unfounded extrapolations [1].

To formalize the internal mechanisms of this cognitive effort, the resolution process is divided into phases of knowledge acquisition and knowledge application (already introduced in Section 1.2). In an opaque environment, the solver does not initially possess the rules governing the system. Therefore, the acquisition phase requires intense exploratory activity to identify the logical structure of the problem, culminating in the construction of a mental model. However, as emphasized by Funke (2012), drawing on the pioneering studies of Dörner (1986), this phase does not consist of the indiscriminate collection of data, but demands a rigorous and continuous reduction of irrelevant data [4]. Indeed, the structural limitations of human working memory make it impossible to cognitively process the entire volume of variables in a highly interconnected system. The efficient solver must therefore implement a strategic selection of information, learning to isolate crucial signals from the “background noise” generated by environmental complexity.

Once a mental model has been constructed, however imperfect or partial, the application phase begins, during which the individual attempts to initiate a “dynamic decision making” process to guide the system toward the desired state [4]. It is in this instance that intransparency and dynamics exert the greatest stress on executive functions. Because the system evolves over time and reacts to received inputs, the solver’s actions inevitably alter the environment, rapidly rendering newly acquired knowledge obsolete. Dörner and Funke (2017) identify the failure to manage this stress as the so-called “emergency reaction of the intellectual system” [1]. Under the pressure of complexity, inexperienced solvers tend to exhibit a marked deterioration in performance: “actors tend to reduce their intellectual level by decreasing self-reflections [...], they show a tendency for fast action with increased readiness for risk, [...] and they degenerate their hypotheses formation by construction of more global hypotheses and reduced tests of hypotheses” [1].

Compounding this framework of potential overload is polytely, meaning the presence of multiple goals that are often in direct conflict with one another. In real-world scenarios, optimizing one variable frequently means compromising another, generating decision dilemmas that are not exclusively logical or mathematical, but deeply strategic and ethical. This obstacle prevents the application of single linear heuristics, such as the VOTAT (*Vary One Thing At a Time*) strategy, which is only useful for simple and isolated problems, forcing the individual to instead accept a constant systemic compromise [1].

To manage such a level of conflict and tolerate the frustration deriving from uncertainty, CPS cannot be configured as a single, isolated act of resolution, but must be structured as a cycle prolonged over time. Fischer, Greiff, and Funke (2012) [9], cited as a theoretical foundation by both Dörner and Funke (2017) [1] and Gyaurov, Fabricatore, and Bottino (2025) [3], define CPS as a process of

iterative self-regulation. It requires the subject to engage sequentially in identifying anomalies, continuously formulating and selecting hypotheses, and executing action plans, constantly calibrating their behavior in response to incomplete and uncertain information. By virtue of this dynamic, solutions to complex problems are rarely unique or perfect; rather, they take the form of a strategic assemblage, resulting from “creative combinations of knowledge and a broad set of strategies” [1].

It is precisely the erratic, emotional, and iterative nature of CPS that exposes its deep incompatibility with the paradigms of traditional educational systems. Formal learning environments are historically structured to train students in solving toy problems – problems artificially isolated from their systemic context and stripped of background noise. Such exercises guarantee deterministic success provided the taught algorithm is blindly applied, but they preclude the development of tolerance for ambiguity. The absence of specific training aimed at curbing the cognitive “emergency reaction” leaves the individual completely unprepared when facing the complexity of the real world. Consequently, there is an urgent need to identify new educational media capable of simulating risk, cascading consequences, and information overload, triggering systemic learning within a protected yet intrinsically motivating environment.

2.2 Cognitive and Motivational Mechanisms of Gameplay Learning

If, as analyzed in the previous section, CPS imposes an overload on working memory and triggers emotional stress dynamics, it becomes necessary to understand through which mechanisms interactive systems manage to mitigate this load, fostering the acquisition of systemic skills. To overcome the superficial dichotomy between formal learning and mere pastime, the literature offers rigorous psychological and motivational interpretations, which are essential for decoding the true cognitive potential of the video game medium.

A first fundamental contribution to explaining the effectiveness of video games in managing complexity derives from the studies on gameplay learning conducted by Gee (2003) [10]. According to the author, video games operate as problem-solving spaces in which knowledge is never presented in an abstract, symbolic, or decontextualized form. Conversely, gameplay learning is intrinsically situated: the system’s rules, cause-and-effect relationships, and the information necessary to advance are discovered by the player solely through direct interaction and empirical observation. This continuous contextualization serves as cognitive scaffolding, enabling the user to process networks of interconnected variables that, if presented in a static environment, would be intellectually inaccessible.

In addition to contextualization, video game systems excel in a fundamental

operation for mitigating the stress of CPS: distributed knowledge. As highlighted by Gee’s subsequent work (2005) [11], in video games, intelligence and information do not reside exclusively in the user’s mind, but are distributed between the player and the digital artifact. Through continuous feedback loops (visual, auditory, narrative) and an interface that dynamically tracks the state of variables (resources, time, relationships), the game acts as an extension of the player’s working memory. This unburdening allows the brain to redirect its executive resources from the mnemonic storage of data toward high-level strategy formulation and the prediction of future scenarios.

However, exposure to a situated and intransparent environment inevitably entails the possibility of making mistakes. In real-world scenarios, an error often triggers the aforementioned “emergency reaction” (Section 2.1). In video games, conversely, the game architecture performs a fundamental psychological restructuring of failure. Granic, Lobel, and Engels (2014) [12] emphasize how video games promote significant emotional resilience, transforming failure from a punitive and definitive outcome into a necessary epistemic tool. In gameplay, a flawed hypothesis or a disastrous choice does not represent the end of the cognitive process, but rather generates a productive failure: a game over or a negative narrative consequence provides the feedback needed to calibrate the user’s mental model for the next attempt.

To fully understand what prevents the player from abandoning the system in the face of repeated failures, it is necessary to integrate the emotional dimension. Plass, Homer, and Kinzer (2015) [13] postulate that game-based learning is founded on a three-dimensional architecture capable of simultaneously triggering three forms of engagement: cognitive, affective, and motivational. It is precisely the synergy between narrative immersion (affective engagement) and the gratification derived from understanding the system (motivational engagement) that provides the mental resources required to sustain the high effort demanded by problem solving.

This intrinsic propensity for persistence finds its most solid formal explanation in Self-Determination Theory (SDT), applied to game design by the studies of Przybylski, Rigby, and Ryan (2010) [14]. The authors demonstrate that high-yield game environments attract the user because they manage to satisfy two innate psychological needs of human beings: the need for competence and the need for autonomy. In a complex system, the freedom to test different hypotheses, explore different narrative branches, and experience personalized consequences deeply satisfies the need for autonomy. In parallel, the game’s feedback loop, which rewards strategic adaptation, satisfies the need for competence [14]. It is this constant psychological fulfillment that transforms the frustration of uncertainty into a powerful exploratory engine.

In light of these dynamics, the concept of meaning-making, already introduced in Section 1.2 as the primary source of intrinsic motivation, assumes a structural

connotation. As theorized by Fabricatore (2018) [5], the player does not merely passively decode the proposed narrative, but engages in the logical co-creation of the game world. In an environment characterized by intransparency, the causal architecture of events is never provided in its entirety. It is the user who must fill the information gaps intentionally left by the designer, formulating hypotheses about the underlying rules, the long-term consequences of their choices, and the true intentions of non-playable characters. This active construction of the mental model transforms the solver from a mere executor of inputs into a veritable co-author of the logical ecosystem: the simulated reality acquires a definitive and coherent form only through the interpretive and strategic filter of the player.

For this delicate process to be triggered, however, game mechanics and cognitive challenges must be deeply intertwined. Corroborating this, the meta-analysis conducted by Wouters et al. (2013) [15] highlights how the educational effectiveness of video games drops drastically if the instructional component is simply overlaid onto the entertainment mechanics, rather than being their foundational core. In other words, solving the complex problem must never represent an artificial interruption of the experience, but must perfectly coincide with the gameplay itself.

Such empirical evidence demonstrates that training problem-solving skills under conditions of uncertainty does not require the rigid design of purely educational simulators. Instead, it demands the architecture of ludic ecosystems in which the strategic concealment of information, risk management, and decision-making freedom merge into a holistic and intrinsically motivating experience.

2.3 The Methodological Framework: IMS and CPS-GFC

As argued in the previous sections, for a video game to act as a cognitive training ground for problem solving, the design of uncertainty must be structural and not accidental. To operationalize this necessity by translating psychological constructs into concrete game design tools, this work adopts the methodological framework proposed by Gyaurov, Fabricatore, and Bottino (2025) [3].

This study proposes an integrated approach that combines a game-centric perspective with a player-centric one. To do so, the authors employ two complementary theoretical lenses: Work System Theory (WST) and Cognitive Work Analysis (CWA). The objective of this hybridization, as stated by the authors themselves, is to “systematically extract, code, and interpret gameplay elements and mechanics in terms of psychological processes relevant to CPS” [3].

Rather than rigidly translating these engineering theories, the framework dynamically adapts them to the video game context. WST is employed to model gameplay as a goal-oriented system, in which the player is framed as an operator using specific

tools and knowledge to overcome challenges. Through this lens, game interactions are analyzed to understand the extent to which various elements might enable or hinder task execution [3]. Concurrently, CWA provides the methodological rigor to map how the game environment supports or requires the activation of the cognitive processes necessary to manage uncertainty, without imposing a rigid procedurality on the player's actions.

From the joint application of these analyses, the framework distills two strictly interdependent design tools: Information Management Strategies (IMS) and Complex Problem Solving - Gameplay Feature Configurations (CPS-GFC).

2.3.1 Information Management Strategies (IMS)

IMS describe the design strategies used to generate or mitigate uncertainty by selectively acting on the flow of information available to the player. As formalized by Gyaurov et al. [3], these strategies operate by revealing or concealing details related to four fundamental dimensions of an uncertain phenomenon: its presence, its causes, its effects, and the rules that govern it. The goal of this informational modulation is to induce the player to enact specific cognitive processes, such as the perception of ambiguity, hypothesis formulation, strategic selection, or direct field testing. The framework classifies seven strategies:

1. **IMS1:** Revealing or concealing information regarding the *presence* of an uncertain phenomenon to enable or require players to engage in uncertainty management processes concerning the *presence* of a second uncertain phenomenon.
2. **IMS2:** Revealing or concealing information regarding the *causes* of an uncertain phenomenon to enable or require players to engage in uncertainty management processes concerning the *causes* of a second uncertain phenomenon.
3. **IMS3:** Revealing or concealing information regarding the *effects* of an uncertain phenomenon to enable or require players to engage in uncertainty management processes concerning the *effects* of a second uncertain phenomenon.
4. **IMS4:** Revealing or concealing information regarding the *rules* of an uncertain phenomenon to enable or require players to engage in uncertainty management processes concerning the *rules* of a second uncertain phenomenon.
5. **IMS5:** Revealing or concealing information regarding the *causes* of an uncertain phenomenon to enable or require players to engage in uncertainty management processes concerning the *effects* and *rules* of that phenomenon.

6. **IMS6:** Revealing or concealing information regarding the *effects* of an uncertain phenomenon to enable or require players to engage in uncertainty management processes concerning the *causes* and *rules* of that phenomenon.
7. **IMS7:** Revealing or concealing information regarding the *rules* of an uncertain phenomenon to enable or require players to engage in uncertainty management processes concerning the *causes* and *effects* of that phenomenon.

2.3.2 Complex Problem Solving - Game Feature Configurations (CPS-GFC)

While IMS describe information modulation at a theoretical level, the CPS-GFC represent their practical translation within game mechanics. The framework identifies ten core configurations, conceived as building blocks for game design [3]. Each of these is specifically coded to satisfy two complementary requirements: one aimed at ensuring Player Engagement (PE) and one oriented toward stimulating the development of cognitive skills (Complex Problem Solving - CPS) through IMS.

1. Persistent game world:

PE: Should provide display of the inventory, screenshots of significant events, signs of battles on the avatar, and persistent effects of player actions on the game world.

CPS: Should conceal or reveal the rules of changes to the inventory, avatar state, or persistent game world.

2. Unexpected Game Events:

PE: Should provide schemas that govern player interactions and unexpected game events.

CPS: Should conceal or reveal the causes of the relationship between the unexpected local and global game events.

3. Unpredictable characters:

PE: Should provide reasonable, visible, consistent entities with unpredictable behaviors.

CPS: Should conceal or reveal the effects of unpredictable behaviors of entities.

4. Game goals:

PE: Should provide clear and explorable links in the nonlinear hierarchy of goals.

CPS: Should reveal the rules of exploring and backtracking.

5. Guiding game environment:

PE: Should provide a target direction based on the shape of levels, placement

of collectibles, presence of enemies, and environmental changes.

CPS: Should reveal the presence of collectibles, enemies, and environmental changes.

6. Safe planning:

PE: Should provide undisturbed time in an identifiable protection area based on the patterns of nearby hazards, enemies or environments.

CPS: Should reveal the rules of undisturbed time in protection area.

7. Unfamiliar challenges:

PE: Should provide new and harder challenges by combining familiar game elements in unfamiliar ways.

CPS: Should reveal the rules of familiar game elements.

8. Prospective learning:

PE: Should provide a game mechanic that, once learned, will be used in a future challenge.

CPS: Should reveal the rules of a previously introduced game mechanic.

9. Risk-reward:

PE: Should provide conditional access to safe pathways with small rewards and dangerous pathways with big rewards.

CPS: Should reveal the rules of accessibility and safety of pathways.

10. Warning information flows:

PE: Should provide an unavoidable challenge of obvious difficulty with accompanying audio and visual cues.

CPS: Should provide the presence of intensive audio and visual cues.

The structural integration of IMS within the ten CPS-GFC configurations provides, in summary, a formal and comprehensive design vocabulary. As outlined in this section, the framework does not merely theorize cognitive processes but offers operational tools to modulate uncertainty while simultaneously safeguarding player engagement. Having defined this theoretical and methodological arsenal, it now becomes possible to employ it as an objective analytical lens. The following section will therefore examine some commercial video games to empirically demonstrate how the complex problem-solving dynamics just described already find effective, albeit often implicit, application within the video game industry.

2.4 Analysis of Commercial Titles

Having defined the methodological foundation and the CPS-GFC framework [3], this chapter employs these tools as an analytical lens to examine a selection

of three commercial video games: *This War of Mine* (2.4.1), *Papers, Please* (2.4.2), and *Frostpunk* (2.4.3). The objective of this investigation is to empirically demonstrate how CPS dynamics already find rigorous and mature application within the entertainment industry. Each title will be broken down to highlight the complex scenario it proposes, validating the analysis through the precise mapping of game mechanics onto the ten operational configurations (CPS-GFC) and supporting these observations with the relevant scientific literature.

2.4.1 This War of Mine (11 bit studios, 2014)

This War of Mine [16] subverts the traditional paradigm of war-themed video games: the player does not control a soldier, but manages a group of civilians trapped in a city under siege, whose sole objective is to survive until the ceasefire. From the very first moments, the scenario is configured as a logical and moral ecosystem of extreme complexity, characterized by a drastic scarcity of resources, partial information, and constant threats.

Supporting this intrinsic complexity, the study conducted by Smale, Kors, and Sandovar (2019) [17], based on direct interviews with the designers at 11 bit studios, highlights how the game was conceived to force the user to confront a “wicked problem”. The required choices never possess a univocally correct solution, but rather impose agonizing compromises. The need to scavenge for food or medicine, for example, frequently clashes with the psychological deterioration of the characters, generating a brutal conflict of priorities (polytely) between physical survival and human ethics.

Furthermore, to heighten the sense of disorientation and stimulate the autonomous creation of mental models, the designers deliberately removed any form of traditional tutorial. As the authors emphasize, the game adopts a strict “no hand-holding” policy, in which the user is thrown into the deep end to experience the trauma and uncertainty of the siege [17]. This approach forces the solver to autonomously fill information gaps, investigating the system’s rules through exploration and productive failure.

To systematize this design architecture, Table 2.1 illustrates the detailed mapping of *This War of Mine*’s mechanics onto the ten guidelines of the CPS-GFC framework [3].



Figure 2.1: Game interface of *This War of Mine* during the daytime phase inside the shelter.



Figure 2.2: Game interface of *This War of Mine* showing item crafting using a simple workbench in the early stages of the experience.

Table 2.1: Mapping of *This War of Mine* mechanics onto the CPS-GFC guidelines.

CPS-GFC Configuration		Implementation in <i>This War of Mine</i>
Persistent World	Game	The game adopts permadeath. Every action consumes resources irreversibly, and the characters' psychological or physical traumas persist without the possibility of reloading a previous save.
Unexpected Events	Game	The system generates stochastic shifts, such as the sudden arrival of winter (which requires stoves and wood) or the intensification of fighting that temporarily blocks access to specific districts.
Unpredictable Characters		Civilians or soldiers encountered during exploration or who knock on the shelter's door possess unpredictable alignments. They may offer advantageous trades, beg desperately for help, or set lethal ambushes.
Game Goals		Strong presence of polytely. The macroscopic objective (surviving) is fragmented into perpetually conflicting micro-objectives (e.g., stealing medicine from civilians to save a companion, thereby causing depression within the group).
Guiding Game Environment		Absence of on-screen markers. Visual and auditory clues, such as boarded-up doors, rubble, or the sound of footsteps through ceilings, indirectly guide the player in assessing risks.
Safe Planning		The day/night dichotomy structures time. The daytime phase confines the characters to the shelter, providing a safe zone free from external threats to manage the inventory, build tools, and heal the wounded.
Unfamiliar Challenges		In the early stages, the player confronts novel threats (e.g., a gunshot wound) without any prior tutorial on the bandaging or resting procedures required by the game.
Prospective Learning		The player learns stealth mechanics (e.g., hiding in the shadows or making noise to attract attention) in uninhabited scenarios, and later transfers and applies these tactics in lethal military zones.
Risk-Reward		The night map imposes continuous cost-benefit evaluations. Extremely high-risk locations (e.g., a military outpost) hold resources of vital importance (weapons and bandages), contrasting with safe but resource-poor areas.
Warning Information Flows		Building a makeshift radio acts as a vector for broadcasting local news, allowing the player to partially anticipate upcoming cold snaps or price hikes for goods on the black market.

2.4.2 Papers, Please (Lucas Pope, 2013)

Papers, Please [18] casts the player in the role of a border inspector for the fictional regime of Arstotzka. Unlike most commercial titles, the gameplay unfolds entirely through a two-dimensional interface simulating the inspector’s desk, where the user must inspect, cross-reference, and stamp the documents of travelers seeking asylum, work, or transit.

The scenario acts as an excellent simulator of cognitive overload and problem solving under time pressure. Validating this setup is the study by Formosa, Ryan, and Staines (2016) [19], who highlight how the game adopts a systemic approach to generate complexity. Instead of relying on predetermined narrative sequences, the title generates ethical and cognitive dilemmas solely through its rigid rules.

As analyzed by the authors, the game forces the user to manage a highly polytelic structure, fragmented into three constantly conflicting objectives: obedience to the State’s bureaucratic directives (which change and become more complex daily), empathy toward the pleas of migrants, and, above all, the survival of one’s own family. The inspector is paid on a piece-rate basis for every correctly processed traveler; however, the continuous updates to the procedures require increasingly more time for the visual analysis of discrepancies [19]. The user thus finds themselves trapped in a continuous trade-off between speed of execution (necessary to earn the vital credits to feed their family) and accuracy (essential to avoid financial penalties from the State).

This closed and oppressive environment represents an ideal case study for the CPS-GFC framework [3], as it demonstrates how intransparency need not necessarily derive from spatial exploration, but can emerge from the sheer volume and ambiguity of the provided information. Table 2.2 illustrates in detail how the title’s mechanics operationalize the ten design configurations.



Figure 2.3: Game interface of *Papers, Please* during the phase of accepting or rejecting a traveler.

Table 2.2: Mapping of *Papers, Please* mechanics onto the CPS-GFC guidelines.

CPS-GFC Configuration	Implementation in <i>Papers, Please</i>
Persistent Game World	The health status, finances, and basic needs of the player’s family persist from day to day. Fines accumulated for inspection errors irreversibly impact the household economy.
Unexpected Game Events	Sudden terrorist attacks at the border drastically interrupt the work shift, halting the influx of travelers and limiting the expected daily income.
Unpredictable Characters	Travelers present high uncertainty: they may lie, use forged documents, offer bribes, or hide contraband, compelling the player to maintain constant investigative skepticism.
Game Goals	Conflict between bureaucratic efficiency (to earn money), adherence to Arstotzka’s rules (to avoid penalties), and personal morality (helping a refugee at the risk of one’s own safety).
Guiding Game Environment	The limited desk space and the physically flippable rulebook implicitly guide and constrain the spatial organization of information. The player must structure their workspace to optimize visual analysis.
Safe Planning	At the end of each work shift, the evening summary screen provides a <i>Safe planning</i> environment. Free from time limits, the player manages savings by critically allocating them among rent, food, and medical care.
Unfamiliar Challenges	The game constantly introduces new investigation tools (e.g., X-ray scanners, fingerprint detectors) that the user must quickly learn to master and integrate into their workflow.
Prospective Learning	The ability to spot basic discrepancies, learned in the early levels, is the prerequisite for resolving the complex webs of forgeries presented in the advanced stages.
Risk-Reward	Favoring the rebel organization or accepting bribes from smugglers offers high monetary rewards but entails the lethal risk of arrest for treason and a subsequent game over.
Warning Information Flows	The morning bulletin provides anticipatory information, alerting the player to dangerous fugitives, epidemics (requiring vaccination certificates), or imminent diplomatic retaliations, allowing them to recalibrate their focus.

2.4.3 Frostpunk (11 bit studios, 2018)

Frostpunk [20] expands the concept of survival, taking it to a macroscopic scale. The title hybridizes city-builder mechanics with those of survival management games, casting the player as the Captain, the leader of the last city on Earth during a global ice age in an alternative nineteenth century. The entire settlement is built around a massive coal-powered generator, the sole source of heat and hope for the community.

The presented scenario is a perfect ecosystem of interconnected systems where uncertainty is the driving force of the experience. The study conducted by Troost et al. (2023) [21] specifically examines how *Frostpunk* masterfully models uncertainty and decision-making processes in response to extreme climate crises. The authors highlight how the core of the game revolves around constant strategic ambiguity: the player is continually forced to ask themselves “What should I do?” when faced with ethical and management dilemmas devoid of painless solutions.

Polytely, in this context, is institutionalized through the *Book of Laws*. Indeed, the player must balance resource management and efficiency (such as coal and the workforce) with the need to take a stance on fundamental ethical issues, often having to choose between industrial optimization and human well-being [21]. The introduction of child labor or lethal emergency shifts increases productivity, but drastically raises Discontent and plummets Hope – the two persistent indicators that, if taken to extremes, result in the player’s exile and a game over.

This network of systemic interdependencies requires a continuous process of CPS. Table 2.3 demonstrates how *Frostpunk*’s mechanics map perfectly onto the configurations of the CPS-GFC framework [3], forcing the player to devise long-term forecasts in a ruthless environment.



Figure 2.4: Game interface of *Frostpunk* during the panoramic view of the city.

Table 2.3: Mapping of *Frostpunk* mechanics onto the CPS-GFC guidelines.

CPS-GFC Configuration		Implementation in <i>Frostpunk</i>
Persistent World	Game	The promulgated laws are permanent and irreversibly alter the social fabric. Levels of Hope and Discontent persist and govern the player’s political survival.
Unexpected Events	Game	Sudden temperature drops, unexpected arrivals of sick refugees, or the discovery of other fallen cities drastically alter management priorities and the thermal and economic balance of the settlement.
Unpredictable Characters		The population reacts emergently to living conditions. Citizens may organize strikes, form dissident factions, or make specific demands (e.g., heating homes) based on the uncertainty of the moment.
Game Goals		Profound conflict between industrial expansion and maintaining the humanity and morale of the population, forcing the constant acceptance of compromises.
Guiding Game Environment		The crater map and the radial thermal radiation of the generator implicitly guide urban planning: the player visually deduces that vital and residential structures must occupy the inner, warmer rings.
Safe Planning		Periods of temporary climatic stability and the ability to pause time provide safe spaces in which to plan upcoming moves (e.g., constructing buildings, allocating the workforce).
Unfamiliar Challenges		Citizens’ demands and the final Great Storm, which presents temperatures so low as to render previous heating strategies obsolete, compel the player to devise new management strategies.
Prospective Learning		The thermal micro-crises faced in the early stages serve as an essential cognitive foundation: the player learns to manage coal stockpiles and medical care to prepare for the final disturbance.
Risk-Reward		Sending scout teams into the frozen wastelands requires evaluating whether to push further to recover industrial relics and survivors, at the risk of losing the entire team to weather events.
Warning Information Flows		The weather interface provides accurate forecasts of impending temperature fluctuations, allowing the player to anticipate the load on the generator and initiate research for new thermal insulation in time.

Conclusions of the Analysis

The analysis conducted on related works has allowed for the empirical validation of the theoretical and methodological assumptions presented throughout this chapter. The examination of *This War of Mine*, *Papers, Please*, and *Frostpunk* has demonstrated how game mechanics, when modeled to simulate the typical attributes of a complex system, can elevate an entertainment product to a CPS simulator.

Through the analytical lens of the CPS-GFC framework [3], it has clearly emerged how cognitive overload and uncertainty represent deliberate and necessary design tools. They force the player to abandon the passive execution of procedures to embrace systemic investigation, hypothesis formulation, and the assumption of responsibility in the face of irreversible ethical dilemmas.

Having consolidated this robust methodological foundation, the discussion can now proceed from the analytical phase to the synthetic and implementation one. In the following chapter, the ten operational configurations discussed and validated will serve as actual engineering foundations for the design and development of the original prototype that is the subject of this thesis. These guidelines will no longer be employed to deconstruct existing works, but rather to architect a new interactive experience capable of operationalizing uncertainty and cognitive load within an original ludic environment.

Chapter 3

Design and Implementation

This chapter illustrates in detail the design and development process of the serious game that is the subject of this thesis. The primary objective of this digital artifact is the operational translation of the CPS-GFC theoretical framework [3], described in Chapter 2, into concrete game mechanics, system architectures, and narrative structures. Every design choice, from the conceptualization of the theme and plot to the implementation of the software architecture, was guided by the need to stimulate the cognitive processes associated with CPS. The following sections will analyze the initial concept, the transposition of the guidelines into game mechanics, the system architecture, and the integration of the narrative engine.

3.1 Game Vision and Concept

The project materializes as a narrative adventure with a text-based interface, integrated with basic selection and interaction mechanics. The concept of the work revolves around the figure of a soldier left isolated in hostile territory following an armed conflict. Deprived of the support of their chain of command, lacking communications, and immersed in a radically foreign environment, the protagonist must embark on a journey to return home.

The shaping of the project into a text-based experience, devoid of explicit spatial or environmental exploration, does not stem from the assumption that such a format offers intrinsic cognitive or pedagogical advantages for learning CPS compared to a freely navigable virtual world. On the contrary, the adoption of this genre is the result of a careful feasibility analysis and an engineering compromise between the structural requirements imposed by the CPS-GFC framework and the stringent constraints of project management.

The reference framework requires the design of an ecosystem dense with interconnections, capable of generating uncertainty, long-term consequences, and

branching decision trees. Translating systemic complexity of this magnitude into a traditional 3D or 2D gaming environment would have entailed development times incompatible with the lifecycle of a thesis project. Furthermore, it would have required the mobilization of technical-artistic skills (3D modeling, animation, spatial level design) falling outside the primary objectives of this research and difficult for a single developer to sustain.

Therefore, the purely text-based interface was identified as the most efficient vehicle to ensure the realization of the work. This project scoping choice allowed almost all time resources to be allocated to the pure engineering of the game logic, the architecture of the branching narrative, and the correct implementation of the mechanics necessary to stimulate Complex Problem Solving, entirely bypassing the production bottlenecks linked to the creation of advanced graphic assets.

3.1.1 Setting and Systemic Uncertainty

The decision to adopt a war setting and to focus the narrative on an isolated military protagonist is neither purely aesthetic nor oriented toward entertainment for its own sake. Conversely, it specifically addresses the pedagogical needs imposed by the CPS-GFC framework.

A scenario of isolation in enemy territory represents, by its very nature, the pinnacle of systemic uncertainty. The player is immersed in a context where information is structurally incomplete or unreliable: they do not know the exact morphology of the territory, are unaware of the deployment of hostile forces, and cannot predict with certainty the long-term consequences of their actions.

This condition of information deprivation forces the user to abandon problem-solving approaches based on pure motor execution or instinctive reactions, compelling them instead to activate higher-order cognitive processes. The player must gather fragments of information, formulate hypotheses about the state of the world, and test these hypotheses through cautious choices — an iterative process that constitutes the core of CPS learning.

3.1.2 Deconstructing “The Enemy” and Polytely

A cornerstone of the game vision is the deconstruction of the traditional concept of the enemy. In commercial war-themed video games, the antagonist is typically a monolithic entity, clearly defined and recognizable, with whom interaction often concludes through combat or elimination dynamics. In this serious game, however, hostility becomes an opaque, nuanced, and multidimensional entity.

The NPCs populating the plot, such as the figures of Elias and Lira, are not mere pawns but embody complex social and psychological dynamics. They possess personal motivations, fears, and agendas that are not explicitly exposed to the

player. This social opacity forces the user to suspend heuristic judgment (e.g., wearing an enemy uniform therefore means being a lethal threat) to evaluate each encounter critically and contextually.

From the perspective of the CPS framework, this relational ambiguity serves to generate and support polytely (see Section 1.1). The player is subjected to the constant tension between their primary goal of pragmatic survival (which might suggest ruthless actions to save resources or eliminate potential threats) and the maintenance of their own ethical or relational integrity. Navigating this gray area, weighing decisions that never present an overtly correct solution free of negative externalities, is a fundamental exercise for the development of systemic and critical thinking.

3.1.3 Core Loop

To translate these theoretical ambitions into a playable experience, the foundational infrastructure was organized around a strictly text-based core loop (the fundamental iterative cycle of interaction), inspired by the branching structures of gamebooks and interactive fiction. The macroscopic objective of the experience – escaping the hostile territory – provides the long-term motivational drive, but actual progression relies on the repetition of a cycle of abstraction and choice.

The core loop is articulated into four sequential phases based exclusively on the processing of written information:

1. **Information Acquisition:** The player reads and analyzes the textual descriptions provided by the current narrative node, critically extrapolating salient data about the environment, the characters, and the contingent situation.
2. **Systemic Evaluation:** The player is required to monitor the current state of their explicit resources (Health and Fatigue), evaluating imminent needs and attempting to deduce, solely through the cues provided by the text, the state of social relationships.
3. **Decision Making:** The user selects an option from the proposed textual branches (which may represent physical actions, dialogue choices, or tactical approaches), often operating under conditions of profound temporal or informational uncertainty.
4. **Feedback Assimilation:** The system returns the consequences of the action through a new block of text, simultaneously updating narrative variables and numerical resources. This forces the player to readjust their mental map to face the subsequent node.

This read-evaluate-choose cycle is strongly constrained by a system of statistics operating on two different levels of transparency, requiring the player to act under a regime of *Safe Planning*. On one hand, the interface provides quantitative feedback on visible resources such as Health (physical integrity) and Fatigue (the energy expendable to overcome obstacles), preventing purely instinctive or physically unsustainable choices.

On the other hand, the system tracks latent variables behind the scenes, such as Cohesion (psychological and relational resilience within the group). This statistic influences certain narrative passages and the attitude of companions toward the protagonist, but it is not visible to the user. This precise design choice serves to deliberately increase the system’s intransparency: in the absence of explicit numerical indicators, the cognitive load shifts entirely onto textual interpretation, compelling the player to mentally simulate the social consequences of their actions.

In summary, the game concept acts as a microcosm generating complex textual problems, wherein the player learns to navigate the unknown by balancing systemic risk and social consequences to achieve their goals.

3.2 Translating the CPS-GFC Framework into Game Mechanics

The game design foundation of the project was established with the primary objective of translating the theoretical guidelines for CPS and IMS [3] into a system of game rules and mechanics. During this conceptualization phase, typical game design vocabulary was integrated into a rigorous structure, where gameplay features are designed to balance the cognitive demand resulting from the selective concealment of certain information with the support provided by revealing others.

As a text-based adventure with a management component, the game intentionally omits the navigation of a physical space (such as a map or a three-dimensional level). Instead of classic environmental exploration, the experience requires the user to navigate a network of narrative branches and rigid resource constraints. In this context, challenges do not rely on reflexes or motor skills, but rather on deduction, risk management, and the ability to construct constantly updatable mental models. Table 3.1 summarizes the conceptual mapping between the theoretical framework and the core features devised for the game. In the following paragraphs, this translation is analyzed detailing its underlying design rationale.

CPS Guideline	Implementation in the Game
Persistent Game World	Design of a closed and not automatically reversible economy: resources are not restored, and vital variables can only be regenerated through the use of resources or player choices.
Unexpected Game Events	Introduction of branches and unforeseen events that disrupt the resource equilibrium, invalidating the player's short- and long-term plans.
Unpredictable Characters	Presence of interactions with characters whose behavioral outcomes elude the player's direct control.
Prospective Learning	The system fosters the internalization of systemic and ethical rules to be applied in heterogeneous narrative contexts.
Game Goals	Fragmentation of the final goal into emerging and shifting sub-goals, dictated by the state of necessity.
Risk-Reward	Actions with high potential for material reward always expose the player to a latent risk concerning vital or relational variables.
Guiding Game Environment	Replacement of spatial navigation with logical exploration: the environment provides clues in the form of text and alterations to status parameters.
Warning Information Flows	Design of an indirect feedback system that signals the approach to critical thresholds.
Safe Planning	Alternation of ludonarrative phases with phases designed purely for status and inventory evaluation and the restoration of physical variables (the camp).
Unfamiliar Challenges	Progressive introduction of situations that disrupt strategies based solely on the quantitative optimization of resources.

Table 3.1: Mapping between the CPS-GFC guidelines and the serious game mechanics.

3.2.1 World Persistence

The first design challenge was the translation of the *Persistent Game World* guideline. To stimulate complex reasoning, the player must perceive that their actions carry a concrete and definitive weight. Therefore, the game was designed around a particularly rigid resource economy. Classic mechanics of video game entertainment, such as automatic health regeneration or an abundance of healing items, were discarded. In their place, a system was devised in which indicators such as *Health* and *Fatigue* undergo reversible variations solely through the consumption of resources or specific player choices, effectively establishing a polytelic triad consisting of keeping physical variables below the critical threshold, optimizing resources, and narrative choices.

This design forces the user to abandon a tactical and reactive approach in favor of a long-term strategy. Directly tied to this mechanic is the *Safe Planning* guideline. The design of the game's pacing entails the alternation between moments of high cognitive demand and moments of mechanical decompression. During these safe phases, the game introduces no new threats, but instead requires the player to interrogate the state of their resources, analyze the inventory, and plan their next moves, providing the necessary time to process previously gathered uncertain information.

3.2.2 Risk and Objective Structure

In the CPS framework, uncertainty is a fundamental component. At the mechanical level, this translates into a continuous manipulation of *Game Goals*, *Risk-Reward*, and *Unexpected Game Events*. Although the macroscopic objective of the game is clear from the outset, the design ensures that it is not sufficient to guide moment-to-moment choices, proposing emergent goals that influence the player's strategy.

The design of risk fits into this context. The choices offered to the player are not dichotomies between a right and a wrong option, but rather complex balancing acts where IMS are used to conceal the exact cause-and-effect relationships. The player can evaluate whether to risk increasing their *Fatigue* to explore an area in search of vital resources, accepting the uncertainty of a possible unforeseen event (*Unexpected Game Events*). The inclusion of disruptive events in the design does not serve a punitive purpose, but aims to test the player's mental flexibility, compelling the rapid recalibration of a strategy that is no longer optimal.

3.2.3 Environmental Complexity

The absence of a three-dimensional graphical representation poses an intrinsic structural limit to the implementation of the *Guiding Game Environment* and *Warning Information Flows* guidelines. In traditional video games, level design

heavily leverages the show, don't tell principle, using lighting, spatial geometry, and visual landmarks to silently guide the player's intuition. In a text-based adventure, where the medium almost entirely coincides with explicit storytelling, replicating this tacit guidance represents a complex design challenge, as a truly visible and explorable environment is lacking.

To overcome this limitation, the environment was conceptually de-spatialized. The design dictates that exploration does not occur by moving an avatar, but by decoding information: the user interface and the text effectively assume the role of the physical world. In the absence of actual spatial environmental mechanics, the system relies on narrative text and parametric textual feedback. The deliberately nuanced textual descriptions and the variations in status indicators act as indirect sensors. The design never provides the exact success probability of an action or the precise nature of an impending danger; instead, it reveals only the existence of an anomaly. The player must rely on their own deduction to connect the dots between the critical state of their parameters and narrative hints, actively filling the information gaps left deliberately open by the designers to stimulate hypothesis formulation.

3.2.4 Social Complexity and Prospective Learning

The implementation of interpersonal dynamics (*Unpredictable Characters* and *Unfamiliar Challenges*) required addressing a structural challenge inherent to the text adventure genre itself: the stimulation of *Prospective Learning*. In a traditional spatial video game, prospective learning is mechanical and visual: the player can tackle a simple puzzle in a preliminary phase and apply the same skills in a more complex environment later. The affordances (e.g., a specific lever or physical obstacle) remain unchanged and easily recognizable, ensuring the transfer of acquired skills.

Conversely, in a purely text- and narrative-based game, the overlap between mechanic and story makes this transposition problematic. Repurposing a narrative situation explicitly linked to an already experienced event solely to test a mechanic would generate strong redundancy, undermining immersion. On the other hand, excessively masking the problem behind an entirely original narrative situation would hinder the player's ability to recognize similarities and apply the learned heuristic.

To overcome this medium limitation, the design abstracted *Prospective Learning* from the situational level, elevating it to the systemic-social level. One of the ways to generate uncertainty consists of concealing the rules governing NPCs' reactions to the protagonist's decisions. Consequently, the player is not called upon to learn how to repeatedly overcome a specific physical obstacle, but rather to progressively internalize the ethical and social rules of the game. Transferring the mental models

learned during initial social interactions to resolve novel dilemmas in the final stages represents a good compromise to convey the narrative's central theme to the user and integrate *Prospective Learning* into a text-driven medium.

3.3 Narrative Design and Experience Structuring

Section 3.2 illustrated the translation of the CPS guidelines into game mechanics, defining the systemic infrastructure and the resource economy. However, for these rules to generate actual cognitive demands, they require an application context. Narrative design serves precisely this function: it transforms mechanics into concrete situations and dilemmas that the player can tackle. In a text-based adventure, narrative is not a mere aesthetic device, but represents the very medium through which the dynamics of uncertainty are conveyed to the user. This section analyzes the structuring of the gaming experience, starting from the central theme and then proceeding to a detailed analysis of the narrative scenarios and the game sub-situations of which they are composed, illustrating how the plot translates the IMS and the guidelines of the CPS-GFC framework [3] into playable situations.

3.3.1 The Central Theme of the Story

As outlined in Section 3.1.2, the purely narrative framework of the serious game is based on the deconstruction of the psychological distance between the protagonist and the enemy. The macroscopic objective is to return home through a devastated territory. The war remains in the background as an element of constant pressure, while the narrative design aims to stimulate profound reflection through encounters and situational crossroads. Instead of forcing a predetermined moral, the system strategically distributes clues (such as Elias's incomprehensible notebook or the encounter with the hermit) to provide the player with the cognitive tools necessary to overcome the threat heuristic, granting them total decision-making autonomy and the freedom to draw their own conclusions about the human complexity of the characters encountered.

3.3.2 The Plot as a Sequence of Problem Spaces

The narrative plot has been structured as a concatenated sequence of problem spaces, in which each node represents a specific configuration of uncertainty. The experience opens with a premise that establishes the protagonist's isolation and the scarcity of resources, immediately setting the paradigm of the *Persistent Game World*.

The narrative branches into interconnected situations. Each of these was designed to provide the player with partial information, requiring the application

of decision-making strategies. The pacing of the experience alternates high-tension phases (e.g., exploration of hostile areas or encounters with NPCs) with *Safe Planning* phases (the map screen or the camp scenario), in which the pace slows down to allow the player to evaluate their inventory and physical condition, as well as to organize and consume resources.

Specifically, the narrative progression unfolds along a path that culminates in a macroscopic central bifurcation. The structural map of the experience can be divided into four main phases:

- **Introductory Phase and First Social Stand-off:** The journey begins in a ruined *Farm*, where the player becomes familiar with the exploration mechanics and faces the first ethical dilemma upon meeting Elias, a wounded enemy deserter. If the choice is made to abandon him, the player obtains his notebook, written in the incomprehensible language of the enemy. Fleeing from a patrol, the action moves to the *Woods*, the scene of a high-tension stand-off with Lira, an undercover allied spy. In this phase, the group of companions who will face the rest of the journey is defined.
- **The Radio Puzzle:** Upon reaching a burned *Village*, the pace slows down. The player must explore the ruins to gather the components necessary to operate an old military radio. The resolution of this problem space provides the tactical information essential for tackling the game's major narrative crossroads.
- **Mutually Exclusive Paths:** Depending on their condition and the information gathered, the player chooses one of the two routes to reach the border. The *Military Road* requires a stealthy approach among enemy convoys and the management of alert mechanics; the *Mountain Pass*, conversely, forces a brutal exercise in survival against nature and toxic gas residue, wearing down physical variables. In both paths, if Elias is not with the protagonist, the retrieval of a bilingual dictionary and the subsequent translation serve as the thematic pivot for the humanization of the adversary.
- **Thematic Resolution:** All choices converge at the *Border*. This final phase presents no complex systemic challenges, but validates the player's assimilation of the narrative's theme. Depending on how many companions have survived, the game culminates in a final act of compassion toward a trapped young enemy soldier, or in the bitter realization of one's own dehumanization reflected in the face of an anonymous corpse in the mud.

We will analyze each scenario in detail in Subsection 3.3.4.

3.3.3 Failure Management: Collapse Events and the Absence of Game Over

In structuring the experience, a crucial aspect of narrative design concerns the management of failure. In a traditional video game adventure, the depletion of vital resources (typically reaching a value of zero for the *Health* statistic) coincides with the irreversible interruption of the playthrough (game over). However, this conventional solution is incompatible with the nature and objectives of a serious game designed for research: the premature interruption of testing would preclude the collection of valuable data regarding the continuation of the user's decision-making experience.

To resolve this methodological limitation, the system replaces the game over with the implementation of *collapse events*. When the *Fatigue* statistic reaches the critical threshold of 100, the game forces a narrative interruption describing the protagonist's physical breakdown and loss of strength. At the systemic level, this event translates into a severe but non-terminal penalty:

- If the protagonist is traveling with companions, they will be forced to intervene to bring him to safety or act in his stead, causing a decrease in the *Cohesion* variable.
- In certain situations, the collapse will result in a loss of material resources.

Following the event, *Fatigue* is forcibly restored to a lower value (typically 60), allowing the player to resume the adventure. This device establishes a symbolic failure that punishes the imprudent management of resources and is carefully tracked by the system for the subsequent data analysis phases. From this perspective, the *Health* statistic is also configured as a pressure variable: it exists to ensure immersive coherence and incentivize the player to weigh their tactical choices, even though it cannot trigger irreversible consequences that would halt the experience.

3.3.4 Analysis of Game Situations

The main ludonarrative situations that make up the journey are analyzed in detail below. For each of them, the salient points of the mapping to the theoretical framework and the rationale behind their design are highlighted.

The Camp – Strategically placed between the various scenarios, this isolated narrative hub acts as a moment for recovery and decompression.

- **Implemented Guidelines:**
 - *Safe Planning*: Environmental pressure is temporarily suspended, allowing the player to evaluate their inventory and plan future moves without time constraints.
 - *Persistent Game World*: Recovery actions (consuming rations, resting) alter global statistics (*Fatigue*, *Health*), making them fundamental for preparing for the rest of the adventure.
 - **Implemented IMS:** The node does not apply information concealment or revelation strategies, but serves as a threat-free environment necessary for processing the information gathered in previous nodes.
 - **Design Rationale:** The user is required to perform purely managerial operations (e.g., consuming a ration to increase *Health* and sleep better, thereby decreasing *Fatigue*). Furthermore, it offers the opportunity to converse with companions, delving into their backgrounds and increasing *Cohesion*.
1. **The Farm** – This opening scenario introduces the player to exploration mechanics and their first moral dilemma.
 - (a) **Initial Exploration** – The protagonist wakes up wounded on the floor of a ruined kitchen.
 - **Implemented Guidelines:**
 - *Persistent Game World*: The choice to explore the surrounding environment increases the *Fatigue* statistic, instructing the user from the very first moments about action economy.
 - *Warning Information Flows*: Textual details (smoke on the horizon, fresh tracks) act as indirect environmental warnings.
 - **Implemented IMS:**
 - *IMS1*: The game provides visual fragments that enable the deduction of the area’s danger level and the probable presence of nearby enemies.
 - **Design Rationale:** The cognitive demand forces the player not to move blindly, but to formulate hypotheses regarding the location’s safety by processing descriptive texts as actual danger sensors.
 - (b) **The Unstable Pantry** – The player identifies an unstable area that could contain fundamental resources.
 - **Implemented Guidelines:**

- *Risk-Reward*: Direct opposition between the possibility of obtaining a highly useful resource (the crowbar) and the probability of suffering physical damage.
 - **Implemented IMS:**
 - *IMS3*: The text immediately clarifies that attempting to retrieve the resources could compromise the building’s structural integrity, requiring the player to plan their moves by hypothesizing the effects of a potential collapse (e.g., loss of resources, physical damage).
 - **Design Rationale:** The situation requires the user to rationally evaluate whether the long-term value of the resource outweighs the immediate physical risk.
- (c) **The Encounter with Elias** – The discovery of an enemy soldier, wounded and unarmed.
- **Implemented Guidelines:**
 - *Unpredictable Characters*: The enemy NPC does not exhibit predictable behavior; his reactions to aid or hostility are not guaranteed.
 - *Unfamiliar Challenges*: The player is challenged to make a choice based on empathy or mistrust, without relying on pure logic.
 - **Implemented IMS:**
 - *IMS6*: The effect of interaction choices with Elias is concealed, requiring the player to hypothesize the causes of his condition and how he might behave (rules).
 - **Design Rationale:** The absence of clear metrics forces the player to make a decision based solely on their ethical compass, introducing the theme of humanizing the enemy. If the player decides not to help Elias, they will obtain his notebook, written in the enemy’s language and currently incomprehensible. Later, they will have the opportunity to translate a portion of it.
2. **The Woods** – The player is intercepted by Lira, an undercover agent from their own faction who considers them a threat.
- **Implemented Guidelines:**
 - *Unpredictable Characters*: Lira’s attitude changes dynamically based on what the player says or does, and her mistrust scales up if Elias is present.
 - *Warning Information Flows*: The text signals the woman’s growing nervousness, warning of the imminent exhaustion of her patience.

- *Unfamiliar Challenges*: The player finds themselves at a distinct tactical disadvantage and has no choice but to play along with Lira while seeking a way to expose her, in order to avoid surrendering precious resources.
- *Unexpected Game Events*: The protagonist’s journey is unexpectedly interrupted by stepping on a landmine, causing the situation to evolve rapidly and forcing the player to confront a problem they were unprepared for.

- **Implemented IMS:**

- *IMS2*: By revealing that the mine belongs to the protagonist’s army, the player understands it is not an improvised trap but a standard military device, allowing them to deduce that without an expert bomb disposal technician, DIY defusal is almost impossible.
- *IMS5*: Concealing the cause of Lira’s behavioral dissonance requires the player to formulate hypotheses about the interaction rules to apply with her.
- *IMS7*: Revealing Lira’s operational rule allows the player to predict the effects of the interaction and discard hypotheses regarding negative outcomes.

- **Design Rationale:** The interaction is not a simple dialogue tree, but a veritable timed puzzle driven by hidden state variables. The player is given the opportunity to expose Lira by gathering clues.

However, the system introduces a mechanic of attrition: every observation or question consumes the NPC’s *Patience* variable. The player must balance gathering evidence (*Accusation Strength*) and maintaining an appropriate tone (*Respect*) before patience runs out. In case of failure – namely, launching a premature accusation or exhausting available attempts – Lira does not engage in a lethal firefight, but instead penalizes the player by seizing vital resources (such as rations or bandages) before vanishing into the woods.

Furthermore, the complexity level of this unmasking is dynamically influenced by prior choices. If the player is accompanied by Elias (who wears an enemy uniform), Lira’s initial mistrust is exacerbated: the mathematical thresholds to convince her become more stringent, and the success rate drops. Conversely, if the protagonist is alone, the deductive process is more forgiving. This variation serves a dual design function: on one hand, it provides coherent narrative reactivity (paranoia toward the enemy); on the other, it acts as an invisible balance, offering a player left without allies a slightly easier opportunity to gain a new traveling companion.

3. **The Burned Village** – Upon reaching the devastated settlement, the pace of action slows down. The player must explore the ruins in a non-linear fashion to gather the components necessary to operate an old military radio.

(a) **The Underground Shelter** – Access to a basement is blocked by heavy rubble, forcing the player to evaluate the energy expenditure required to enter.

- **Implemented Guidelines:**

- *Risk-Reward*: The player must decide whether to sacrifice precious *Fatigue* reserves (or use the crowbar) to access uncertain resources.
- *Prospective Learning*: Retrieving sensitive documents will provide knowledge applicable in subsequent phases.

- **Implemented IMS:**

- *IMS1*: The system reveals the presence of bodies and documents inside the shelter, allowing the player to manage strategic planning and deduce the usefulness of the loot.

- **Design Rationale:** The challenge requires a purely managerial cost-benefit evaluation for access. Once inside, the narrative adapts dynamically: alongside the note containing the radio frequency (necessary for the next node), the presence of companions opens up crucial systemic bifurcations. If Lira is part of the group, she will recognize the corpse of an allied deserter, enabling the discovery of the *Mountain Pass* map. This event is not merely narrative, but effectively unlocks an entire macro-area of the game that would otherwise be inaccessible, expanding the player's decision-making space.

(b) **The Merchant in the Barn** – An encounter with an armed scavenger possessing the radio decryption manual.

- **Implemented Guidelines:**

- *Unpredictable Characters*: The scavenger poses an obstacle to obtaining an important resource, and the limit of his patience is not explicit.

- **Implemented IMS:**

- *IMS6*: The effect of the interaction with the merchant is concealed, forcing the user to test the effectiveness of bartering or threatening based on their own resources.

- **Design Rationale:** This node imposes a choice of social approach. The player can decide to yield resources to buy the manual (if they have anything to barter) or attempt intimidation. However, the system silently evaluates the group's strength: a threat will only succeed if

backed by numerical superiority or firearms; otherwise, it will result in failure and a potential confrontation.

- (c) **The Command House and the Radio** – The use of the central machinery to intercept communications.

- **Implemented Guidelines:**

- *Unfamiliar Challenges:* The calibration puzzle deviates from the survival logic seen previously.

- **Implemented IMS:**

- *IMS7:* Proceeding randomly without knowing the radio tuning rules risks breaking it. The player needs external information (frequency and manual) to solve the problem safely.

- **Design Rationale:** This sub-situation constitutes the core of the area. Initially, the radio is unusable. The player must decide how much to invest in exploration before testing the system: attempting a blind calibration leads to the definitive breakage of the device, precluding access to vital information. Trying with only the frequency is difficult, but not impossible. Possessing both the frequency and the manual, however, allows the challenge to be overcome safely.

- (d) **The Strategic Crossroads** – Based on the information obtained from the radio, the player must choose between two mutually exclusive scenarios. A successful use of the radio unlocks specific logical options that mitigate uncertainty, while still keeping the challenge non-trivial.

- **Implemented Guidelines:**

- *Guiding Game Environment:* The text summarizes the learned information in a briefing useful for guiding the player toward the choice they find most appropriate given their current state.
- *Safe Planning:* The player can freely reflect on the available information and plan their next moves without rushing.
- *Game Goals:* The player is called upon to choose the path to follow, identifying the next goal from two possible choices.

- **Implemented IMS:**

- *IMS7:* By revealing (totally or partially, depending on the outcome of the radio calibration) the conditions of the two paths, the player must formulate a hypothesis regarding which one is more sustainable for their current state.

4. **The Military Road** – The player faces an environment patrolled by enemies. If they decrypted the transmissions via the radio in the village, they know the guard change schedule, unlocking a strategic waiting option that reduces

initial uncertainty. Once past the introductory phase, the core of the level imposes a fundamental crossroads.

- (a) **The Warehouse Route** – Proceeding through the old depots offers partial cover but introduces dangerous proximity to the troops.

- **Implemented Guidelines:**

- *Guiding Game Environment:* Textual descriptions focus on environmental details (e.g., the presence of enemies, specific containers with a peculiar appearance), providing the player with the necessary information to visualize the protagonist’s situation.
- *Unexpected Game Events:* Making mistakes increases the hidden *Alert* variable, which culminates in the unexpected event of being discovered by a patrol, forcing an immediate route recalibration. Furthermore, the truck breakdown constitutes an unforeseen event that forces the player to update their strategy.
- *Risk-Reward:* The player must weigh the opportunity to obtain valuable loot by searching the containers against the high risk of triggering the enemy alert.

- **Implemented IMS:**

- *IMS4:* The guards’ alert level is not a numerical variable accessible to the player, who must therefore carefully evaluate how to move to avoid being discovered.
- *IMS5:* By revealing the reason for a vehicle’s stop (a mechanical breakdown), the player must deduce the new rules governing the situation. The fact that the soldiers are stationary invalidates the previous plan and requires adapting one’s strategy to the new situation.

- **Design Rationale:** The warehouses offer potential loot but introduce a hidden *Alert* mechanic. While crossing, the protagonist comes across a container whose appearance catches their attention, leading the player to evaluate whether to try to enter to recover potential resources (a smoke grenade) or avoid the risk of alerting the guards. Subsequently, the protagonist encounters a truck that stops due to a mechanical breakdown, attracting enemy soldiers. The player must decide whether to use the smoke grenade to cover their passage, proceed stealthily, or exploit the soldiers’ distraction. If the guards’ alert reaches the limit, the player is forced into a premature and costly escape.

- (b) **The Drainage Channel** – Descending into the drainage channel represents the tactical alternative to the warehouse route, prioritizing total concealment at the expense of physical endurance.

- **Implemented Guidelines:**
 - *Risk-Reward*: Trading the tactical safety guaranteed by concealment for a severe and mathematical drain on the *Fatigue* statistic due to the extreme environment.
 - *Warning Information Flows*: Textual descriptions warn the player of enemy movements and the imminent stopping of the vehicle exactly above their position.
 - **Implemented IMS:**
 - *IMS5*: By revealing the reason for a vehicle's stop (a mechanical breakdown), the player must deduce the new rules governing the situation. The fact that the soldiers are stationary invalidates the previous plan and requires adapting one's strategy to the new situation.
 - **Design Rationale:** The channel eliminates the risk of being visually discovered but heavily drains *Fatigue* due to the mud and freezing cold. The situation degenerates when a truck suffers a breakdown and stops exactly above the grate where the player is hiding. The player must choose between two risky approaches: remain motionless in the freezing water waiting for the soldiers to leave (suffering an extreme *Fatigue* penalty) or attempt to force the grate to escape, drastically increasing the risk of an alert.
- (c) **The Escape and the Notebook** – The protagonist is discovered by enemy soldiers and forced into a desperate escape. Subsequently, they reach a safe zone in the woods, where the drop in tension gives way to a thematic revelation.
- **Implemented Guidelines:**
 - *Prospective Learning*: Translating the notebook requires the player to assimilate a new ethical perspective, updating their mental model and influencing how they will perceive hostile soldiers in subsequent scenarios.
 - **Implemented IMS:**
 - *IMS3*: If the player is discovered prematurely, the effect of their actions and the guards' alert level is revealed, causing the escape situation. The player must then hypothesize the effects of their next choices to survive.
 - *IMS4*: By revealing Elias's intimate thoughts, the user is predisposed to manage cognitive dissonance and update the mental rule by which they identify the enemy.
 - **Design Rationale:** The section is divided into two consecutive phases. Initially, the player is forced into a desperate flight toward

the woods, during which they can use certain resources to cover their desperate sprint (smoke grenade, pistol, and bullets), and their fatigue can increase up to the critical threshold, leading to a collapse. In this situation, *Cohesion* dictates how Elias, if present, will lose his life. If this variable is sufficiently high, he will sacrifice himself to cover the protagonist's retreat; otherwise, he will separate from them during the escape and be struck by the pursuers' bullets.

Upon reaching the safe zone, and during the moment of rest, a highly specific interaction is unlocked. If the player obtained the bilingual dictionary by stealing it from the container or recovering it in the drainage channel and possesses Elias's notebook (obtained at the beginning of the game at the farm, if Elias did not join the group), they can translate it autonomously. Providing no mechanical advantage for survival, this narrative micro-event serves solely to break down the psychological distance, challenging the user to recognize the profound humanity of those they are trying to evade.

5. **The Mountain Pass** – Choosing the high-altitude route, the player faces an extreme and dehumanizing environment, battered by freezing wind and characterized by ruined infrastructure. Unlike the military road, uncertainty here is not dictated by enemy presence, but by the attrition of physical resources.

- (a) **The Miners' Outpost** – The protagonist reaches a structure whose interior is partially blocked by a cave-in. Beyond the rubble, anomalous wooden crates of military make can be glimpsed.

- **Implemented Guidelines:**

- *Prospective Learning*: If the player decrypted the radio message in the village, they can easily hypothesize that the military crates conceal vital equipment to survive the gas.
- *Risk-Reward*: The player must evaluate whether to invest their own energy or exploit equipment (crowbar) and companions to overcome a physical obstacle.
- *Unpredictable Characters*: The discovery of a single life-saving resource in the presence of multiple allies triggers reactions of trust or suspicion based on the current level of *Cohesion*.

- **Implemented IMS:**

- *IMS1*: Revealing the presence of military equipment crates in an abandoned civilian outpost allows the player to question and formulate hypotheses about their contents.

- **Design Rationale:** The node requires solving a logistics and fatigue problem. To reach the crates, the player must overcome the cave-in by choosing the best approach: using a tool (minimizing effort), asking companions for help (increasing *Cohesion* and limiting *Fatigue* consumption), or forcing it barehanded (suffering a severe physical penalty). Once the obstacle is overcome, opening the crates triggers a dilemma: there is only one functional gas mask. If the protagonist is in a group, the system queries the hidden *Cohesion* variable to determine the NPCs' reactions. High cohesion generates a logical and resigned acceptance of the situation by Elias and Lira, while low cohesion triggers suspicion and fear, setting the psychological stage for the dramatic survival decisions in the upcoming tunnel.
- (b) **The Suspension Bridge** – Before delving into the bowels of the mountain, the player must cross a gorge via what remains of a bridge.
- **Implemented Guidelines:**
 - *Prospective Learning:* Having listened to and understood the radio communications in the village provides valuable information on structural integrity, warning of the danger of rapid movements or excessive weight.
 - *Warning Information Flows:* The text provides indirect environmental warnings describing the dilapidated state of the structure, the rotten planks, and the intensity of the wind gusts.
 - *Unexpected Game Events:* The sudden collapse of the structure under a companion's weight forces an unforeseen and critical rescue action.
 - *Risk-Reward:* The player is forced to balance their tactical choice, paying for the safety of the crossing in terms of pure *Fatigue* or exposing themselves to the risk of physical trauma and the loss of an ally.
 - **Implemented IMS:**
 - *IMS2:* The text describes the state of the structure and the environmental conditions, providing clues to evaluate the stability of the passage.
 - **Design Rationale:** This physical obstacle requires a complex mechanical and managerial evaluation. Analyzing the structure, the player has the opportunity to use an inventory item (the crowbar) to break the ice and unlock a steel cable, thus enabling a new and safer crossing method.
The scenario forces a choice on how to proceed: crawling (requires extremely high *Fatigue* consumption but guarantees grip), running

(fast but highly unstable), or using the freed cable (balanced energy cost). If Elias is in the group, his movement causes an unexpected collapse. The player must decide whether to attempt to grab him or let him fall to save themselves. The chances of a successful rescue are not random but depend on a hidden mathematical calculation: the system dynamically modifies a *Fatigue* threshold limit based on the previously chosen crossing method (e.g., choosing to run drastically lowers the probability of success). If the *Fatigue* accumulated by the player exceeds this threshold, the grip will inexorably fail, leading to the companion's temporary disappearance and a decrease in *Cohesion*. This situation transforms environmental navigation into a brutal and rigorous survival calculation based on one's vital statistics.

- (c) **The Tunnel and the Gas** – Past the chasm, the path continues inside dark mining tunnels saturated with toxic gas.

- **Implemented Guidelines:**

- *Warning Information Flows*: The description of visual cues such as green smoke in the air acts as an imminent warning before incurring lethal penalties.
- *Risk-Reward*: An extreme challenge where the player must continually decide whether to erode their own vital variables to assist companions who need oxygen.

- **Implemented IMS:**

- *IMS7*: Revealing the presence of a safer high route and a gas-saturated low route to cross the tunnel leads the player to question the best way to handle the situation.

- **Design Rationale:** The crossing represents a complex combinatorial problem where variables multiply: the presence or breakage of the mask, the use of tools to quickly force blocked passages (like the crowbar, which saves vital time), and the presence of any allies. If the player is accompanied by Elias or Lira, the challenge expands: if the player recovered both the crowbar (at the farm) and the gas mask, they can choose to split the group into a team that will brave the toxic gas with the mask and one that will use the crowbar to climb onto an elevated walkway, where the air is breathable. Conversely, if the entire group is in the gas, it will drain the companions' strength, and the protagonist must decide whether to slow down to help them or press on. The scenario thus imposes a balance between the instinct for self-preservation and the ethical maintenance of group *Cohesion*.

- (d) **The Hermit** – Emerging from the tunnels, the player reaches an abandoned structure where they encounter the figure of a cynical and isolated

Hermit.

- **Implemented Guidelines:**

- *Prospective Learning*: Assimilating the hermit’s nihilistic view and his harsh critique prepares the player for the resolution of the narrative experience’s ending.
- *Unfamiliar Challenges*: The player faces an atypical verbal and moral conflict where material resources are not at stake, but rather the ethical convictions that drove them to that point.

- **Implemented IMS:**

- *IMS5*: Through dialogue with the hermit, the causes of his nihilistic vision are partially revealed, fostering player reflection on the war and what it entails.

- **Design Rationale:** The encounter opens with an approach choice: threaten the man at gunpoint or show empty hands to peacefully request hospitality. This decision has an immediate mechanical impact on vital variables, as only a peaceful approach allows the player to receive healing and food (lowering *Fatigue* and restoring *Health*). Subsequently, the hermit poses a purely ethical dilemma, stating that soldiers are nothing more than “variables in an equation that has already been solved”. The verbal confrontation serves as a thematic hinge.

In this context, a fundamental interaction is unlocked: if the player is not accompanied by Elias (abandoned previously) but possesses the notebook recovered at the farm, noticing the hermit’s enemy accent, they can ask him to translate it. The old man reads with total cynicism the soldier’s letters to his daughter Sofie and his desire never to touch a rifle again, then throws the notebook at the protagonist, coldly commenting that the man is now just “worm food”. Unlike the military road (where the decryption is autonomous and intimate), here the humanization of the enemy clashes with the apathy of third parties and triggers a harsh inner realization: the game makes the user realize they have dehumanized a man who only wanted to return home.

6. **The Border** – The border represents the macroscopic objective of the entire experience. Having left behind the convoys of the military road or the toxic fumes of the mountain pass, the player reaches a desolate, rain-lashed no man’s land, a sea of mud dotted with trenches and barbed wire. From a game design perspective, this final section introduces no new complex systemic challenges or logistical puzzles, but serves as a thematic closure. The bifurcation between the two epilogues depends on the systemic resolution of decisions made in

previous nodes (having reached the end with at least one companion or having remained alone), leading to two different epilogues.

- (a) **Ending 1** – Condition in which the player reaches the border accompanied by at least one companion (Elias and/or Lira).

- **Implemented Guidelines:**

- *Game Goals*: Achieving the final goal by facing one last test of cooperation and humanity.
- *Prospective Learning*: Validation or definitive rejection of the empathetic mental model built during the game.

- **Implemented IMS:**

- *IMS6*: By concealing the tactical and practical effects of saving or ignoring the trapped boy, the system forces the player to decide based exclusively on the ethical rules they constructed during the journey.

- **Design Rationale**: One step away from safety, the group comes across a young enemy soldier pinned under a steel beam. A moral standoff ensues that reflects the journey's choices: Lira (if present) aims her weapon to eliminate him following cold military logic, while Elias (if present) shields the boy with his own body. The game confronts the player with one final choice: help him or ignore him. Ignoring the boy guarantees quick passage but causes Elias's disgust (who then abandons the group). Conversely, expending the last remaining *Fatigue* to lift the beam provides no tactical advantage but saves the youth. This action symbolically breaks the ruthless logic of war.

- (b) **Ending 2** – Condition in which the player reaches the border alone (having abandoned or let Elias and Lira die), deprived of empathy.

- **Implemented Guidelines:**

- *Game Goals*: Achieving the macroscopic objective in desperate conditions and isolation.
- *Prospective Learning*: The definitive acquisition of the game's alienating theme.

- **Implemented IMS:**

- *IMS5*: Turning the body to search for resources, the player reveals the cause of the dead man's anonymity (his face covered in mud and blood). This discovery forces the player to understand the final rule of war.

- **Design Rationale**: Finding a corpse in the mud, the protagonist loots it out of pure survival instinct. Turning the body to search it,

they discover the satchel is empty but are forced to stare into the dead man’s glassy eyes. If the player had previously translated Elias’s notebook, the text recalls the letters to his daughter Sofie, linking the dead man’s identity to that broken promise. Subsequently, leaning over a puddle, the protagonist discovers that their own face, covered in blood and mud, is now identical to that of the corpse, leading the player to a final reflection on the theme of dehumanizing the enemy and the perceived distance between “us” and “them”.

3.4 Software Architecture and Systems Engineering

To translate the complex theoretical framework of the game design into an interactive application where the system state and player choices are rigorously tracked by the system, the software engineering required the adoption of a hybrid, modular, and highly decoupled architecture. This section illustrates the technical foundations of the project, focusing on the rationale that guided the software design and how the engineering choices intrinsically support CPS requirements and the silent data collection for empirical experimentation.

The following subsections will analyze in detail the methodological reasons behind the choice of development tools, the architectural paradigms and design patterns adopted, the technical integration between the game engine and the narrative scripting language, and finally, the structuring of the telemetry infrastructure necessary for the qualitative-quantitative analysis of the thesis.

3.4.1 Technology Stack: Unity and Ink

The software architecture of the serious game was designed with the dual objective of supporting a complex game design and ensuring the rigorous empirical data collection necessary for the validation of the research hypotheses. To this end, the selected technology stack is based on the integration of the Unity game engine and the Ink narrative scripting language.

The adoption of Unity as the reference engine is primarily motivated by prior knowledge of the development environment, a factor that reduced the technical learning curve and allowed resources to be concentrated on design and implementation. The Unity infrastructure offers a fast iteration speed, supported by the breadth of its asset store. From the perspective of empirical research, the main methodological advantage lies in Unity’s flexibility in supporting the runtime extraction of statistics for data collection.

Regarding the management of the complex narrative branching, relying exclusively on the game engine's features and the C# language to develop and govern large decision and narrative trees would have significantly prolonged development time. Therefore, the project integrated Ink, a language specifically designed for the development of non-linear interactive narratives. The robustness of this tool is amply demonstrated by its use as an industry standard for narrative titles developed in Unity. Numerous successful commercial games, characterized by complex branching plots and advanced state management, rely on the integration of Unity and Ink: examples include critically acclaimed titles such as the very recent *Esoteric Ebb* (Christoffer Bodegård, 2026) [22], *Overboard!* (inkle, 2021) [23], *Heaven's Vault* (inkle, 2019) [24], and *Sable* (Shedworks, 2021) [25]. Adopting a technology extensively tested in the commercial sector guarantees the stability and reliability required for an experimental research project.

Furthermore, Ink meets three crucial architectural requirements for the game's implementation:

- **Separation of Concerns:** It allows for a clear isolation of the story design and content writing from the application programming, ensuring greater project maintainability.
- **Native state serialization:** The engine keeps track of the entire narrative path explored by the player, a feature of inestimable value for the subsequent export and scientific analysis of behavioral traces.
- **Systemic interoperability:** It offers deep binding mechanisms with the game engine, allowing purely narrative choices to translate instantly into logical and systemic consequences governed by the C# code.

3.4.2 Development Paradigms

The architectural structuring of the serious game was conceived with a rigorous engineering approach to ensure the scalability, maintainability, and flexibility necessary to support the continuous iterations required by the game design. The project fully embraces the component-based paradigm intrinsic to Unity, favoring reusability and behavior composition.

At the core of the infrastructure is a strict application of the principle of separation of concerns. The main logical domains of the application, such as the narrative, player state management, user interface, input acquisition, and metrics collection, are encapsulated in distinct systems with no direct coupling. To facilitate communication between these systems, the application adopts an event-driven programming paradigm. By leveraging native C# events, systems that produce information are decoupled from those that consume it. An emblematic

example is the management of variations in the protagonist's physical state (*Health* and *Fatigue*): modifying a parameter emits an event that the graphical interface passively receives to update itself, without the statistics manager needing to know the existence or structure of the UI.

To solve recurring problems and optimize performance, the software design makes targeted use of specific architectural patterns:

- **Scriptable Object-based architecture:** This pattern was employed to create databases of objects (such as resources and companions), to define the structure of immutable information associated with them, and to decouple the input management module. **Scriptable Objects** act as shared data containers and communication channels independent of the current scene, eliminating hardcoded dependencies.
- **Singleton and PersistentSingleton:** Systems requiring a global, centralized access point were implemented using the Singleton pattern. The project distinguishes between scene-specific managers (which are destroyed to prevent inconsistent states) and persistent managers (**PersistentSingleton**) that survive scene transitions, such as those tasked with the cross-cutting management of the application lifecycle.
- **Observer pattern for the game loop:** The implementation of centralized managers for the update cycle stems from the need to optimize the game engine's performance. Assigning native methods (**Update**, **FixedUpdate**, **LateUpdate**) to numerous individual components is indeed less efficient than managing a single control point. By adopting the observer pattern, scripts requiring continuous updates register with these managers by implementing specific subscription interfaces. The manager itself will then, from within its native Unity cycle, orchestrate and cascade the logic of all subscribed components. This centralized approach offers two fundamental architectural advantages: first, it extends the ability to synchronize with Unity's lifecycle to native C# classes (not bound to the **MonoBehaviour** class); second, it allows subscribers to be organized according to precise priority levels, ensuring absolute and deterministic control over the execution order of instructions within each individual frame.
- **Interfaces and Dependency Inversion:** In accordance with SOLID principles, the C# code heavily utilizes interfaces (such as **IStorage** for containers or **IUpdateObserver** for the game loop) to define clear contracts between systems. High-level modules depend exclusively on these abstractions and never on concrete implementations, ensuring an architecture where individual components (e.g., the inventory system) can be replaced or modified without

breaking the code that uses them. To support this paradigm, a custom utility (`InterfaceReference`) was developed to allow the serialization of interfaces within the Unity inspector, a feature not natively supported by the engine. This generic wrapper allows script references to be assigned directly from the editor based solely on the interfaces they implement. This solution significantly expands the possibilities for creating modular, flexible, and highly maintainable systems.

3.4.3 Integrating Ink into Unity

For the interactive experience to act as a true *Complex Problem Solving* environment, the narrative choices made by the player must not be limited to branching the text; they must translate into systemic and tangible consequences within the game state.

From an architectural perspective, it is fundamental to premise that the native Ink runtime for C# operates completely agnostically with respect to the presentation layer. The narrative engine merely manages the story's state machine, the evaluation of logical conditions, and the advancement of the textual flow, exposing an API that returns only raw data (text strings, tags, or lists of available choices). This total absence of a predefined graphical interface grants the programmer maximum freedom to decide how, where, and when the text should be rendered on screen, fostering absolute decoupling between narrative logic and the visual presentation.

To achieve this interoperability and bridge the gap between raw textual data and the graphics engine, the architecture employs a series of fundamental components that act as a structured bridge. The main contributions of this integration are articulated in the following modules:

- **Flow Orchestration:** Implemented following the Singleton pattern to ensure centralized access, the `StoryManager` script represents the core of narrative execution. It is responsible for instantiating the `Ink.Runtime.Story` object, parsing the narrative tree, and orchestrating the user interface. In addition to managing the progressive display of text and the dynamic presentation of choice nodes, the `StoryManager` exposes native C# events (such as `OnStoryEnter` and `OnStoryExit`) that notify other modules of the beginning and end of a narrative situation, triggering general state transitions without introducing direct coupling.
- **Global State Persistence:** Although Ink autonomously manages the local variables of a single story, the game's systemic structure requires the persistence of shared parameters across the entire decision graph. To resolve this fragmentation, the project adopts a master file named `Globals.ink`, which acts as a centralized registry for diegetic variables. At the application level, the `StoryVariablesRegistry` injects this global state into the Ink runtime

at the start of each narrative fragment; at the end of the sequence, it extracts the updated values and stores them, ensuring absolute and serializable state consistency.

- **Bidirectional Interoperability:** The actual mutation of game variables and interaction with external systems are delegated to a class that maps Unity’s logical routines to the **EXTERNAL** functions declared in the Ink scripts. For the purposes of this discussion, we will not delve into the detailed implementation of individual functions; however, it is crucial to emphasize that the game systems written in C# make advanced use of reflection, pattern matching, and careful type management. This ensures that method calls originating from Ink and their passed parameters – natively encapsulated in generic `Ink.Runtime.Object` types – are correctly decoded, validated, and cast to the corresponding C# types, ensuring safe and robust execution of the game logic.

This highly decoupled paradigm allows the author to write high-level narrative commands that systemically modify the environment (e.g., subtracting resources or updating the value of state variables).

3.4.4 Behavioral Tracking and Telemetry for CPS Validation

The empirical validation of the CPS-GFC framework, a central objective of this research work, requires the ability to objectively measure the player’s behavioral and cognitive responses when faced with complex problems and scenarios of uncertainty. To this end, the system architecture was engineered by establishing a clear structural separation between the data defining the diegetic state of the game world and the silent telemetry metrics used for scientific analysis.

This dichotomy materializes in the implementation of two parallel systems: **GlobalStats** and **RuntimeStats**.

GlobalStats represent narrative variables inherently linked to the gaming experience. They quantitatively model the critical dimensions of the complex system in which the player operates, expressed as parameters such as *Health*, *Fatigue*, and group *Cohesion*. At the implementation level, the **GlobalStatsManager** encapsulates these parameters into reactive objects (such as the **IntStat** class), which ensure values are maintained within valid mathematical ranges and autonomously notify the UI of variations by emitting C# events. This approach provides the player with immediate systemic feedback, stimulating resource management and strategic replanning processes.

Concurrently, the system employs **RuntimeStats** for the analytical tracking of the session. These metrics, completely invisible to the player, record objective

behavioral traces: number of choices made, nodes explored, time spent, and resource utilization. To guarantee maximum performance and avoid affecting the gaming experience, `RuntimeStats` are implemented using low-overhead static arrays indexed by strongly typed enumerations (e.g., `IntRuntimeStat`), which ensure constant access time ($O(1)$) and zero dynamic memory allocations during gameplay.

Analytical tracking extends pervasively into the narrative engine as well. Specific silent telemetry variables have been inserted into the Ink scripts, activating solely to mark the adoption of certain heuristics, the perception of specific affordances, or the resolution of moral dilemmas, capturing decision-making nuances that would be difficult to track using the physics or logic engine alone.

The convergence point of this analytics infrastructure is represented by the `StatsExporter`. At the end of the play session, this component aggregates and serializes the collected data into a structured JSON file. The generated document unifies two essential information domains:

- **StoryStats:** An exact snapshot of the Ink variables (including the `CPS_*` telemetry flags) at the time of conclusion, documenting the systemic outcomes and long-term consequences of the actions taken.
- **GameplayStats:** The quantitative summary of the `RuntimeStats`, describing the observable interactive behavior at the gameplay level.

The resulting output, uniquely saved for each session, provides the raw, scientifically measurable, and analyzable dataset that will form the basis for evaluating the Research Questions addressed in the next chapter.

3.5 User Interface as a Cognitive Tool and Feedback Management

In the absence of a three-dimensional spatial environment typical of action or exploration-based video games, the user interface of a text adventure does not merely overlay the game, but becomes the “physical world” with which the user interacts. In the context of developing this serious game, the interface and its associated feedback system were designed to assume the role of a cognitive extension for the player, acting as one of the tools used for the active modulation of uncertainty. The following paragraphs illustrate how the visual design and software engineering choices were guided by the need to support, and simultaneously challenge, the solver’s decision-making processes.

3.5.1 The Interface as a “Logical Environment” and External Memory

Tackling a complex problem requires the simultaneous processing of multiple interconnected variables, a process that risks rapidly exhausting the user’s working memory resources, generating cognitive overload. From this perspective, the application’s HUD was conceived to act as a veritable external memory. By visually structuring the inventory, the group’s status, and available resources, the interface relieves the player of the burden of mnemonically tracking logistical parameters, allowing them to focus their cognitive energy on the strategic evaluation of narrative choices.

This constant visual frame defines the “logical environment” of the complex system. By providing structured and unchanging access to state information, the UI directly enables the application of the *Safe Planning* guideline outlined in the reference framework. The player can indeed suspend action, consult their current variables, and formulate an action plan based on objective data, mitigating the sense of disorientation derived from a hostile scenario.

3.5.2 Modulating Uncertainty: Explicit and Implicit Feedback

The core of the interface design lies in the calculated dichotomy between which information to clearly show the user and which, conversely, to deliberately conceal. This selection is not random but represents a rigorous application of IMS to modulate systemic uncertainty.

On the one hand, the game uses explicit feedback for physical survival parameters, such as *Health* and *Fatigue*. The transparency of these data provides the player with unequivocal systemic limits and acts as a trigger for *Warning Information Flows*: for example, the *Fatigue* value approaching the critical threshold represents a visual alarm signal that forces the user to update their strategy to minimize physical effort.

On the other hand, the design adopts implicit feedback for intangible and social parameters, such as group *Cohesion* or *Alert*. These values actively operate within the game engine, but their numerical data is intentionally hidden within the user interface. The application of this strategy generates an “information opacity” functional to problem solving: concealing the exact data forces the player to deduce the state of social relationships solely through the narrative’s textual feedback and the characters’ diegetic reactions. The purpose of this mechanic is to stimulate the formulation and testing of hypotheses, forcing the solver into a continuous analytical effort to decode the intentions and behaviors of unpredictable characters.

3.5.3 Interface Engineering

To ensure that the User Experience (UX) fluidly supports the required cognitive load without technical interruptions, the interface was engineered in Unity according to robust and scalable architectures.

The navigation system is based on a stack hierarchy of layers, governed by the `UINavigationController` component. This architecture allows the contextual overlapping of menus, inventory panels, and narrative windows to be managed without ever causing the loss of global context; when the user closes a top layer panel, the system automatically restores the state and interactivity of the underlying layers.

For the structured visualization of complex data, such as the grid of possessed items or companions, the Model-View-Controller (MVC) pattern was adopted. Pure logical data (the Model) is decoupled from its visual representation (the View); with every user action that alters the inventory, the Controller updates the model and forces the dynamic repopulation of the UI panels, guaranteeing consistency and precision.

Finally, the synchronization between textual narrative and visual feedback is ensured by a strict reactive system. Specific components, such as the `UIStatsHandler`, do not continuously poll the game state but merely listen passively to the events emitted by the `GlobalStats` system. In this way, the visual interface is always instantly synchronized with the systemic consequences triggered by narrative choices processed in Ink, offering responsive, high-performance, and narratively immersive feedback.

Chapter 4

Experimental Evaluation and Results Analysis

This chapter outlines the validation methodology adopted to evaluate the effectiveness of the serious game and presents the results of the data analysis. Given the nature of the research and the size of the sample involved, it should be noted from the outset that this investigation is formally configured as an exploratory study. Consequently, the statistical analyses conducted do not claim to provide irrefutable proof of the universal effectiveness of the applied framework; rather, they aim to identify trends, correlations, and significant behaviors useful for validating design choices and laying the groundwork for future, larger-scale investigations.

The chapter is structured as follows: Section 4.1 introduces the testing infrastructure and the measurement tools used. Section 4.2 outlines the sample profiling and defines the initial attitudinal baseline. Section 4.3 presents a descriptive analysis aimed at examining engagement levels, the perception of game mechanics derived from the CPS-GFC framework [3], and the overall appreciation of the design; this evaluation is supported by a factual reconstruction of the players' experience based on telemetric data. Section 4.4 constitutes the analytical core of the chapter, where objective data is cross-referenced with subjective perception through the application of inferential statistical tests, in order to formally answer the Research Questions (RQs) that guide the study. Finally, Section 4.5 discusses the methodological limitations of the study.

4.1 Methodology and Measurement Tools

The experimental setup was specifically designed to evaluate the CPS-GFC framework's effectiveness applied to the text adventure. Data collection aims to answer

five Research Questions investigating the relationship between uncertainty mechanics, learning, and player engagement:

- **RQ1 (Perception, hypotheses, and feedback):** To what extent does the design of uncertainty stimulate players to perceive anomalies, formulate and select predictive hypotheses, refining their strategies based on environmental feedback?
- **RQ2 (Prospective learning):** Does the system foster the acquisition of mental models that are effectively transferred to solve novel problems in the advanced stages of the game?
- **RQ3 (Engagement and agency):** How do emotional involvement and narrative immersion foster the internalization of agency and the assumption of responsibility toward the system's irreversible consequences?
- **RQ4 (Systemic reasoning):** Do players perceive the irreversible consequences of their actions, adopting long-term strategic resource planning?
- **RQ5 (Conflicting goals):** To what extent do ethical dilemmas and relationships with characters interfere with the rational optimization of resources, creating a decision-making conflict?

To minimize external environmental factors and simplify playtesting, the infrastructure relied on unsupervised sessions. During these, a silent background telemetry system, invisible to the user, recorded player choices and relevant statistics. To cross-reference objective metrics with subjective perceptions, telemetric data and questionnaires were linked via the tester's personal identifier before being anonymized for academic discussion.

Alongside telemetric tracking, testers' perceptions were measured using two questionnaires administered at both ends of the session:

- **Pre-test questionnaire:** Completed before playing, it collected demographic data, profiled prior gaming experience, and defined attitudes toward complexity, strategic decisions, and moral dilemmas.
- **Post-test questionnaire:** Administered post-experience, it measured overall engagement, systemic learning, and the perception of uncertainty mechanics, including open-ended questions for qualitative feedback.

Both evaluation tools employ a 7-point Likert scale (where 1 corresponds to "Strongly disagree/Never" and 7 to "Strongly agree/Always"). This scale was selected to obtain an adequate level of granularity to measure the nuanced attitudes and perceptions of the participants.

Given the ordinal nature of the data collected via the Likert scale and the small sample size, descriptive statistics prioritized the use of robust measures such as the median and the interquartile range (IQR). These metrics are also methodologically consistent with the choice to adopt non-parametric inferential tests (Spearman, Mann-Whitney) for the analysis of the previously listed Research Questions.

Explicit reference is made to Appendix A for the full consultation of all items administered to the players. It should also be noted that the notation "Pre/Post X" found throughout the rest of the discussion will refer to question number X of the pre-test or post-test questionnaire.

4.2 Sample Profiling

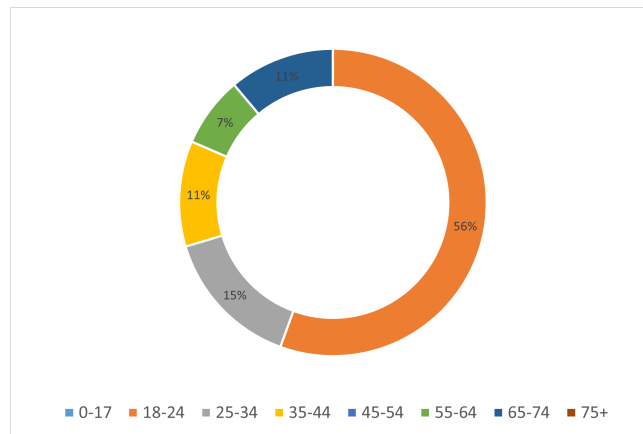
To contextualize the playtest results and evaluate the design's effectiveness, defining the sample's demographics and baseline attitudes is essential. The sample comprises $N = 27$ participants who completed the pre-test questionnaire outlining their prior experience and attitudes toward CPS dynamics and moral choices in video games.

4.2.1 Demographics and Video Gaming Profile

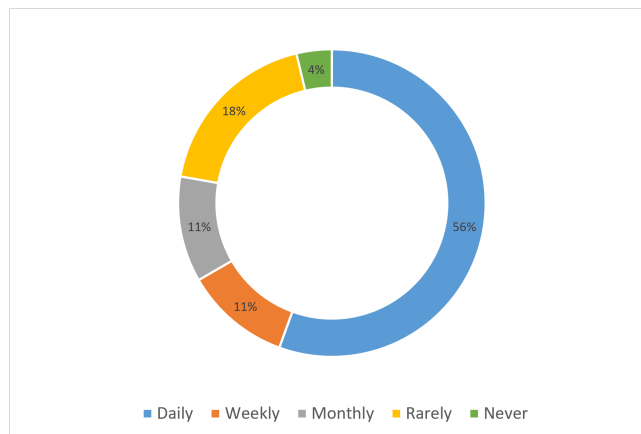
Demographically, the sample spans various age groups (see Figure 4.1a) with a clear prevalence of young adults. Specifically, the majority (15) fall into the 18-24 bracket. The remaining participants are aged 25-34 (4), 35-44 (3), 55-64 (2), and 65-74 (3).

Regarding gaming habits, the data indicates high familiarity with the medium. Over half of the testers (15) play daily. They are joined by weekly (3) and monthly (3) players, compared to a minority of occasional players (5 rarely) and one non-player (see Figure 4.1b).

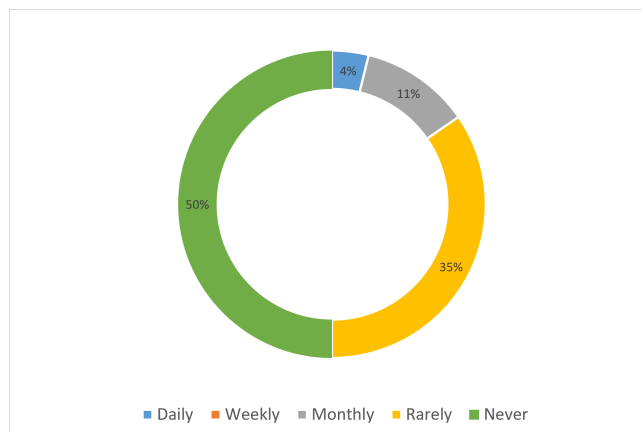
Despite this high general frequency, experience with text-based or narrative-driven titles (e.g., visual novels) is decidedly more limited (see Figure 4.1c). Most play them rarely (9) or never (13), with only 4 showing regular exposure (3 monthly, 1 daily). This limited familiarity is methodologically valuable, allowing the evaluation of the interaction mechanics' accessibility for a non-specialized audience.



(a) Playtesters' age



(b) Gaming frequency



(c) Frequency of playing narrative/text-based titles

Figure 4.1: Demographic and video gaming profiling of the sample

4.2.2 Attitudinal Baseline

The central section of the pre-test questionnaire investigated participants' attitudes toward complex problem solving, uncertainty, and ethical choices. Extracting the medians and interquartile ranges (IQR) allows for outlining the sample's baseline attitudinal profile, which will prove useful for interpreting the telemetric data and post-test responses. Complete results are summarized in Table 4.1.

Table 4.1: Attitudinal baseline of the sample (Values on a 1-7 Likert scale). Refer to Appendix A.1 for the full wording of the items, presented here in a summarized form.

ID	Item Summary (Pre-test)	Median	IQR
Pre 6	I tend to consider long-term consequences rather than focusing solely on the immediate goal.	5.0	2.75
Pre 7	I am interested in understanding the unwritten rules and cause-and-effect systems in video games.	4.5	1.75
Pre 8	I find it stimulating to make morally ambiguous decisions within a video game.	5.5	2.0
Pre 9	Usually, my first reaction to an unknown character is strong mistrust.	4.5	2.0
Pre 10	I prefer having time to plan my strategy before acting.	4.0	3.0
Pre 11	I find it stimulating when a game makes the consequences of my actions irreversible.	4.0	2.75
Pre 12	I enjoy having to interpret hidden clues to anticipate a danger.	5.0	2.0
Pre 13	I prefer interacting with characters who exhibit unpredictable behavior.	4.0	2.0
Pre 14	I tend to sacrifice resources to make morally 'good' choices.	5.0	2.0
Pre 15	I tend to attempt high-risk actions if the potential reward is very high.	5.0	2.0
Pre 16	I like it when unforeseen events disrupt my plans, forcing me to improvise.	4.0	2.0
Pre 17	When faced with a new mechanic, I find it rewarding to deduce the rules autonomously.	5.0	2.75
Pre 18	I usually try to memorize details and patterns, expecting them to be useful in the future.	4.5	3.0

As shown in Table 4.1, the sample exhibits a marked predisposition toward

ethical and decision-making themes, recording the highest median for the item on morally ambiguous choices (Pre 8: Median = 5.5). This aligns with a concrete propensity for tactical sacrifice: testers generally declare themselves willing to yield useful resources for ethically correct actions (Pre 14: Median = 5.0, IQR = 2.0).

From a systemic and strategic perspective, a solid alignment with CPS prerequisites emerges. Players demonstrate a natural inclination toward exploring high-reward risks (Pre 15: Median = 5.0, IQR = 2.0), autonomous interpretation of clues (Pre 12: Median = 5.0, IQR = 2.0), and unguided mechanics (Pre 17: Median = 5.0, IQR = 2.75). Furthermore, a widespread interest in decoding cause-and-effect systems is confirmed (Pre 7: Median = 4.5, IQR = 1.75).

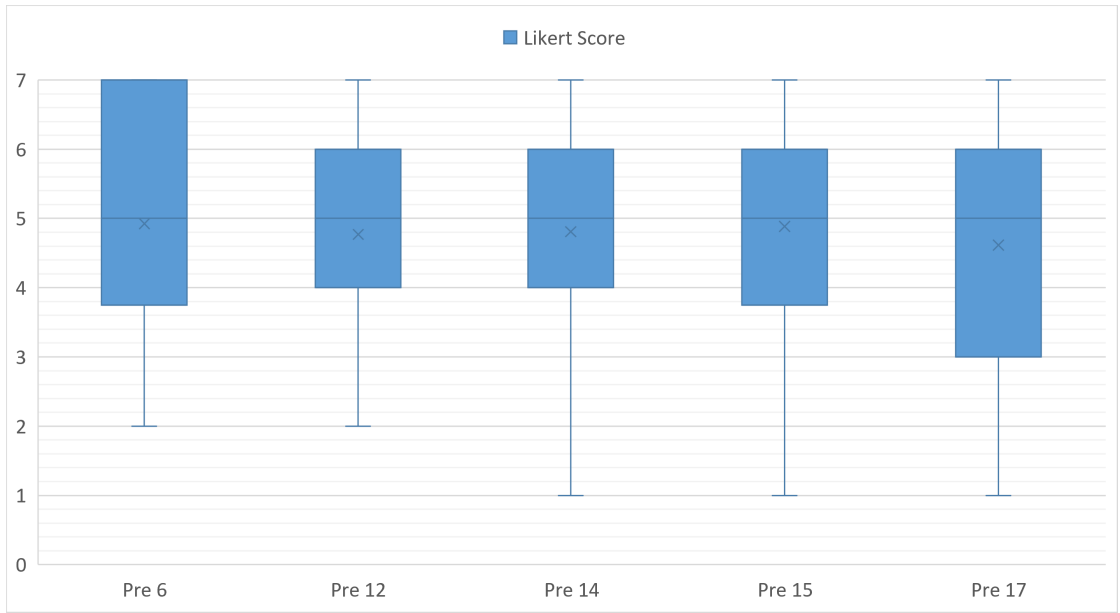


Figure 4.2: Boxplot of the attitudinal profiling results. The graph shows the distribution of scores, expressed on a Likert scale from 1 to 7, for five particularly relevant items from the pre-test questionnaire (see Appendix A.1). Inside the boxes, the horizontal line indicates the median, while the $\times$ symbol represents the arithmetic mean.

It is equally interesting to analyze items with greater data dispersion ($IQR \geq 2.75$), indicating heterogeneous approaches. For example, the dichotomy between careful planning and instinctive reaction divides the sample (Pre 10: Median = 4.0, $IQR = 3.00$), as does the aptitude for memorizing complex behavioral patterns (Pre 18: Median = 4.5, $IQR = 3.00$). Finally, despite a general propensity for long-term planning (Pre 6: Median = 5.0, $IQR = 2.75$), the acceptance of strictly irreversible consequences is more polarized (Pre 11: Median = 4.0, $IQR = 2.75$).

4.3 Experience Evaluation and Telemetric Analysis

This section presents the results from the administration of the post-test questionnaire, cross-referencing the subjective perception of the gaming experience with the objective behavioral metrics collected by the silent telemetry system. Being an exploratory study, the objective is to verify whether the uncertainty dynamics and moral choices designed according to the CPS-GFC guidelines suggested comprehension patterns and triggered systemic behaviors consistent with the initial hypotheses.

4.3.1 Engagement, Immersion, and Perception of Agency

Before investigating the dynamics directly related to CPS, it is useful to evaluate the sample's response in terms of engagement. Post-test data suggests a decidedly high level of engagement and immersion, as visually summarized in Figure 4.3. Despite the absence of traditional graphics, players highly rated the textual descriptions' ability to guide them (Post 21: Median = 7.0, IQR = 1.0) and generate strong emotional involvement (Post 22: Median = 7.0, IQR = 1.0).

Particularly relevant for research purposes is the perception of player agency. Testers report perceiving their choices as significant rather than purely aesthetic (Post 3: Median = 7.0, IQR = 1.0), observing a cause-and-effect relationship between their decisions and the story's evolution (Post 4: Median = 7.0, IQR = 1.0). Equally indicative is the strong consensus that narrative outcomes depended on their actions and not on chance (Post 23: Median = 7.0, IQR = 0.0), supporting the potential effectiveness of the designed environmental feedback (Post 8: Median = 6.0, IQR = 0.0).

4.3.2 Systemic Behavior and Resource Management

A fundamental component of CPS is the ability to analyze a system and plan strategies to mitigate its risks. The subjective perception of the players indicates an activation of these processes, albeit with some variability in the responses: there is a general tendency to exploit moments of calm to reorganize resources (Post 17: Median = 5.0, IQR = 2.0) and to ration items in anticipation of future scenarios, rather than for immediate problems (Post 19: Median = 5.0, IQR = 2.0).

This perception finds factual confirmation when cross-referencing the telemetric data. Analyzing interface usage reveals an interesting behavioral trend (see Figure 4.4a): the propensity to set up a camp (the ludic manifestation of *safe planning*) appears to grow as the frequency of global inventory consultations increases. In turn, this active planning generated tangible consequences on the playthrough. As

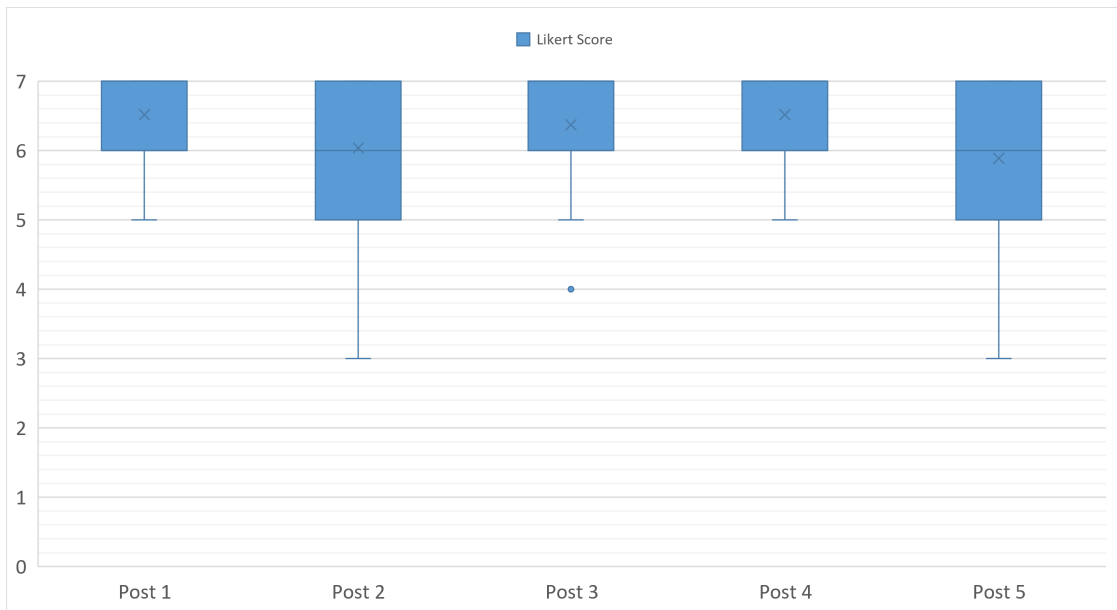


Figure 4.3: Boxplot of responses to the first 5 items of the post-test questionnaire (see Appendix A.2), aimed at evaluating the players' general engagement and their perception of the relevance of the available choices. The graph shows the distribution of scores, expressed on a Likert scale from 1 to 7. Inside the boxes, the horizontal line indicates the median, while the $\times$ symbol represents the arithmetic mean.

illustrated in Figure 4.4b, the collapse trend evidently decreases among those who accessed the camp more frequently. This trend is intrinsic to the game architecture: setting up the base camp allows players to consume resources to rest, lowering their fatigue level before it reaches the critical threshold (100) that triggers an exhaustion collapse.

It is, however, methodologically necessary to report an anomaly that emerged from the telemetry: a substantial portion of the sample (11 players) never accessed the camp during the entire playthrough. Based on informal discussions conducted with testers on the sidelines of the sessions, it is reasonable to hypothesize that this omission did not stem from a conscious strategic choice, but rather from a failure to assimilate the mechanic. The fact that the ability to camp is only mentioned in the text panel introducing the experience, combined with the lack of a sufficiently prominent visual indicator when the first camp is unlocked, likely contributed to users forgetting about it. Nevertheless, the data confirms that players capable of retaining this information and actively exploiting the mechanic were objectively rewarded by the game infrastructure.

A further indicator of the systemic impact of decisions can be observed in the risk-reward mechanics (Post 11: Median = 6.0, IQR = 1.0; Post 20: Median = 6.0, IQR = 1.0). The telemetric data relating to the pantry exploration phase offers an interesting insight in this regard, albeit one to be interpreted with caution due to the strong asymmetry in choices. Among the players who explored the area, the vast majority opted for a more conservative approach, and only a tiny fraction ($n = 3$) chose the risky exploration.

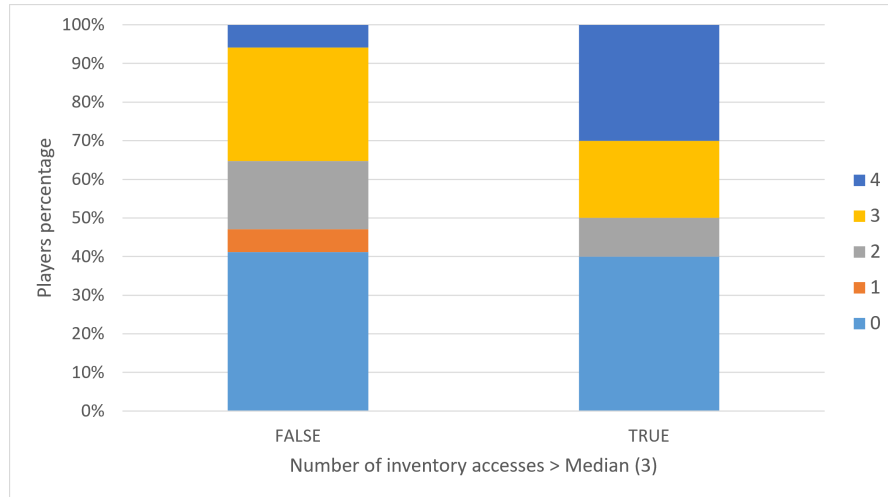
Although this small subsample does not allow for significant deductions, the parameters recorded for these three users are consistent with the design intentions: obtaining a tactical tool (the crowbar) at the cost of immediate physical damage (visible in the drop in minimum health, see Figure 4.5a) allowed them to bypass subsequent obstacles without using brute force, recording lower maximum fatigue peaks than the rest of the sample (see Figure 4.5b).

4.3.3 Moral Dilemmas

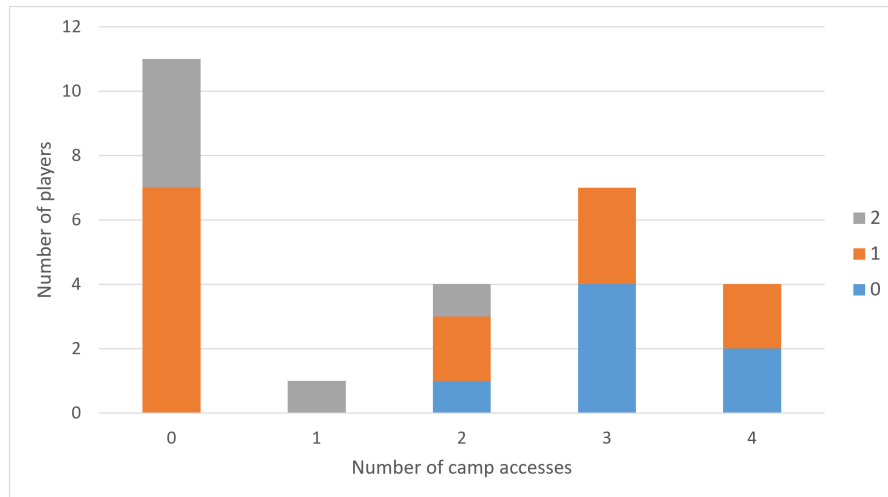
Finally, the design introduced deliberate ambiguities in NPC interactions to stimulate the management of social uncertainty. Players confirmed experiencing difficulty in predicting character reactions, making interactions tense and realistic (Post 14: Median = 6.0, IQR = 1.0).

This degree of uncertainty triggered a conflict between the purely mathematical optimization of resources and adherence to ethical principles (Post 15: Median = 6.0, IQR = 2.0). The game succeeded in its deconstructive intent, leading participants to actively reflect on the humanity of the enemy (Post 24: Median = 7.0, IQR = 1.0). Telemetry confirms the influence of these reflections on practical

choices: cross-analyzing the choice to help vulnerable NPCs (e.g., the boy at the border) against decisions concerning the fate of one’s companions (e.g., Elias) reveals behavioral patterns where empathy frequently prevailed over short-term strategic calculation (see Figure 4.6).

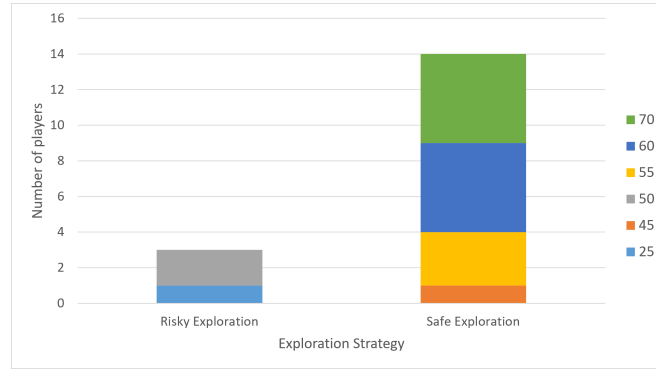


(a) Camp accesses compared to inventory consultations

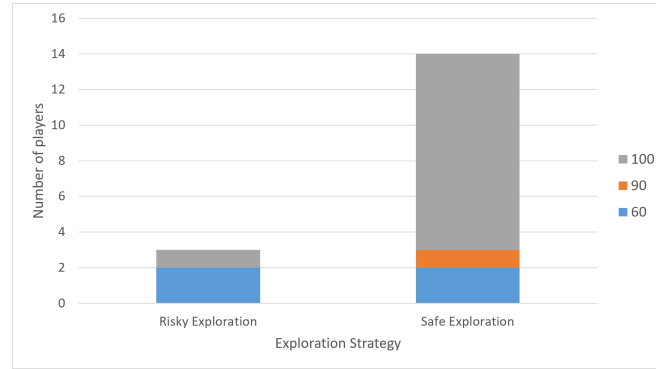


(b) Collapse trend compared to camps accesses

Figure 4.4: Impact of planning on survival performance



(a) Minimum health based on the pantry exploration strategy in the “The Farm” section



(b) Maximum fatigue based on the pantry exploration strategy in the “The Farm” section

Figure 4.5: Systemic consequences of the exploratory approach

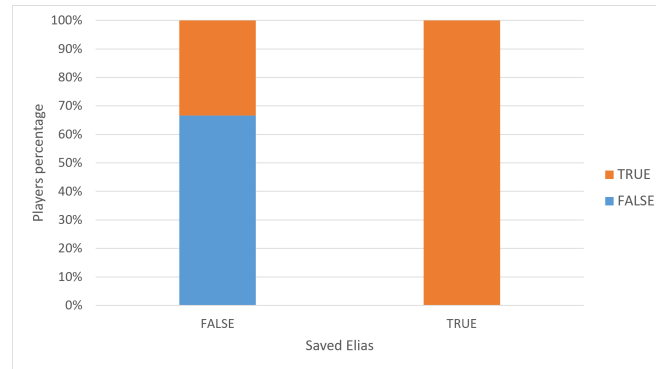


Figure 4.6: Comparison between the decision to help the boy at the border and the prior choice to help Elias (TRUE: the player helped the boy, FALSE: the player did not help the boy)

4.4 Inferential Analysis and Addressing the Research Questions

Having outlined the general trends through descriptive statistics, this section constitutes the empirical core of the research: the resolution of the five previously defined Research Questions (see Section 4.1).

To rigorously investigate the relationships between objective telemetry and subjective perception, non-parametric statistical tests were employed, considered the standard of choice for small sample sizes ($N = 27$) and ordinal variables measured via a Likert scale. Specifically, the strength of linear relationships was measured using Spearman’s rank correlation coefficient (ρ), while the analysis of differences between independent groups was conducted using the Mann-Whitney U test, accompanied by the extraction of the effect size (r) to measure its practical relevance. The significance level for rejecting the null hypothesis was set at $\alpha = 0.05$.

4.4.1 RQ1: Perception, Hypotheses, and Feedback

To what extent does the design of uncertainty stimulate players to perceive anomalies, formulate and select predictive hypotheses, refining their strategies based on environmental feedback?

Data analysis offers an interesting perspective on the management of unforeseen events and the strategic flexibility of players. Spearman’s test revealed a statistically significant negative correlation ($\rho = -0.505$, $p = 0.008$) between the attitudinal propensity to plan meticulously (Pre 6) and the tendency to abandon one’s plans when faced with unexpected obstacles (Post 7).

From a behavioral standpoint, this inverse monotonic correlation indicates that players with a strong planning aptitude demonstrated a tendency to preserve their initial strategy when faced with unexpected environmental feedback. This result suggests a strong retention of the original plan, which can be hypothesized either as evidence of the robustness of the hypotheses formulated upfront by this type of player, or as a physiological cognitive rigidity when facing rapidly changing complex scenarios.

This strategic rigidity, highlighted by quantitative data, finds an interesting qualitative confirmation in the open-ended responses. When participants were asked to describe a moment when they had to alter their plans, some testimonies suggested that adaptation did not occur preventively, but rather manifested as an emergent reaction to the collapse of their initial strategy. The experience described by one user well exemplifies this forced transition:

“When we were discovered by the enemy sentinel at the end of the main

*road, I had to switch from a stealth approach to a desperate escape.” –
Tester 1 (Response to question Post 25)*

Regarding the perception of uncertainty and lack of information, a Mann-Whitney U test was conducted on item Post 9 (the need to proceed with caution in case of incomplete information), dividing the sample between those who solved the radio puzzle and those who did not. The test did not reach statistical significance ($U = 61$, $p = 0.151$, $r = 0.27$). However, observing the descriptive statistics, a slight deviation in the medians is noticeable (7.0 for those who solved the puzzle, 5.0 for the others). Although inferential conclusions cannot be drawn due to the small sample size, this descriptive data allows for hypothesizing a potential trend within the group of testers: those who managed to decode the environmental anomalies might have adopted a more cautious investigative approach. Studies on larger samples would be necessary to formally explore this hypothesis.

4.4.2 RQ2: Prospective Learning

Does the system foster the acquisition of mental models that are effectively transferred to solve novel problems in the advanced stages of the game?

The inferential tests conducted for RQ2 yield the most robust results of the study, strongly supporting the effectiveness of the integrated pedagogical model for those who paid attention to it. The data show a highly significant positive correlation ($\rho = 0.582$, $p = 0.001$) between the self-reported level of reading attention (Pre 2) and the subsequent perception of having learned recurring patterns without explicit explanations (Post 5). This indicates that the players who approached the text with greater meticulousness are the same ones who, in retrospect, reported a stronger acquisition of the system’s opaque rules.

The practical impact of prospective learning finds direct confirmation in objective survival data. A Mann-Whitney U test was conducted on item Post 18 (the ability to deduce solutions for novel challenges based on past clues), dividing the sample between those who completed the play session without ever collapsing and those who suffered at least one collapse. Players who never suffered a collapse reported significantly higher values compared to those who failed, and this difference is highly significant ($U = 18.5$, $p = 0.002$, $r = 0.56$). These findings suggest that players capable of retaining information and formulating accurate mental models enjoyed a tangible systemic advantage.

In further support of the prospective learning process, the open-ended responses offer an interesting insight into how the punitive structure of uncertainty can generate resilient mental models. One tester, for example, explicitly described how an initial negative feedback altered his approach in subsequent challenges:

“I learned, by not going back to the seller of the instruction manual, that without risking or trading other possessions I could have caused more damage (I broke the radio). Therefore, in the next section (the container), I took more risks.” – Tester 4 (Response to question Post 27)

4.4.3 RQ3: Engagement and Agency

How do emotional involvement and narrative immersion foster the internalization of agency and the assumption of responsibility toward the system’s irreversible consequences?

Inferential analysis supports the methodological hypotheses regarding the link between engagement and agency. Spearman’s correlation test revealed a positive and statistically significant relationship ($\rho = 0.418$, $p = 0.030$) between the engagement felt by the participants (Post 22) and the awareness that the evolution of the story depended on their own choices rather than chance (Post 23).

Examining the direction of the correlation, it emerges that as the immersion reported by the individual tester increases, there is a proportional increase in their causal awareness. Players more emotionally involved demonstrated a more solid internalization of agency: having become attached to the narrative, they tended to assume personal responsibility for the plot’s branching, rather than attributing adverse outcomes to the arbitrariness of the game code.

The robustness of this quantitative correlation takes on even greater depth when compared with qualitative data. Open-ended responses provide examples of how narrative immersion pushed some players to formulate complex empathetic reasoning. Emblematic is the case of a user who, despite being aware of the video game’s structural and irreversible limits, voluntarily chose to bear the harshest consequences to protect a virtual companion:

“Since that elasticity [of being able to fix mistakes] is inherently lacking in such a video game, I had to adapt [...]. First of all, I mentally accepted that I would be the first volunteer to sacrifice myself. I would willingly enter a less comfortable situation. This would allow a companion in greater psychological difficulty (Elias) to face the trial to the best of the group’s abilities.” – Tester 20 (Response to question Post 25)

Reflections of this kind suggest how, at least for the players most immersed in the experience, the system might not have been perceived as a mere dispenser of arbitrary penalties, but as a roleplay space in which they came to feel solely responsible for their own fate and that of others.

4.4.4 RQ4: Systemic Reasoning

Do players perceive the irreversible consequences of their actions, adopting long-term strategic resource planning?

Cross-analyzing telemetric data and questionnaire responses aimed to investigate the link between the intention to plan and the actions actually performed in-game. Spearman’s correlation test between the actual frequency of setting up camp and the declared intention to ration resources for future scenarios (Post 19) yielded no statistically significant results ($\rho = 0.149$, $p = 0.458$).

Inferentially, the high p-value indicates insufficient evidence to confirm a linear correlation between rationing intention and the concrete use of rest mechanics. As highlighted in Section 4.3.2, this inconclusive result is likely fundamentally flawed by the behavioral anomaly regarding the failure of 11 testers to assimilate the camp mechanic. This user experience flaw finds explicit confirmation, for example, in a user’s post-game testimony:

“Not that I recall. I certainly didn’t use the information about using the camp that was provided right at the beginning of the experience.” – Tester 13 (Response to question Post 27)

However, integrating this fragmented quantitative data with qualitative post-game testimonies allows for formulating a further exploratory hypothesis. For many users, the concept of “strategic planning” might have translated into pure conservative hoarding out of fear of the unknown, rather than a preventive expenditure to mitigate fatigue. Supporting this interpretation, one tester clearly illustrates this fear of consuming resources, associated with the mechanics’ uncertainty:

“I discovered that resting without eating doesn’t recover much fatigue, which increases the importance of gathering rations (which I didn’t find anyway).” – Tester 9 (Response to question Post 27)

Although inferentially unprovable, this discrepancy between perceived intention and material action offers an important insight for optimizing future systemic feedback.

4.4.5 RQ5: Conflicting Goals

To what extent do ethical dilemmas and relationships with characters interfere with the rational optimization of resources, creating a decision-making conflict?

To measure the friction between survival logic and narrative ethics, a Mann-Whitney U test was conducted on item Post 15 (perceived conflict between conserving

resources and making moral choices), dividing the sample between those who actually chose to save Elias and those who did not.

The test revealed no statistically significant differences between the two groups ($U = 75.5$, $p = 0.636$, $r = 0.09$). The lack of inferential deviations prevents establishing a link between the in-game action performed and the intensity of the perceived moral conflict. However, observing the purely descriptive data reveals that both subgroups settle on a decidedly high median (6.0 on a 7-point Likert scale).

Although generalizable conclusions cannot be drawn, this homogeneity toward the high values of the scale allows for formulating an exploratory hypothesis: it is plausible that the friction between logistical optimization and adherence to empathetic principles generated a cross-cutting decision-making difficulty. This suggests the possibility that the ethical dilemma was perceived by a large part of the sample, regardless of the concrete resolution adopted. As an example, an excerpt from a questionnaire highlights strong immersion, illustrating how the perception of the dilemma pushed the user to voluntarily sacrifice their mechanical advantage:

“Yes. It happened when I chose to provide aid to an enemy by stopping his bleeding. From a strategic point of view, it was a disadvantageous decision, as I preserved a potential threat, but I acted according to the principle of respect for human life, which I consider superior to any factional logic or convenience.” – Tester 7 (Response to question Post 26)

Even in cases where the user favored a pragmatic approach (choosing not to save the companion or the wounded), the weight of the consequences can be tangibly felt, as emerges from another participant’s reflection:

“No, my playthrough was dictated by cynical and cold choices, although I still fully felt the impact of the latter.” – Tester 13 (Response to question Post 26)

4.5 Study Limitations

Although the data analysis yielded encouraging and statistically significant findings regarding the effectiveness of the CPS-GFC framework, it is necessary to highlight the methodological and operational limitations characterizing this investigation. This is to properly contextualize the scope of the obtained results and guide future research directions, reiterating the purely exploratory nature of this study.

First, the size and composition of the sample ($N = 27$) warrant caution regarding external validity and the generalizability of the results. Although the use of non-parametric inferential tests (such as Spearman and Mann-Whitney) ensured the mathematical rigor required for small samples, the investigation involved a

demographically unbalanced population. As highlighted in Section 4.2, the user base is heavily concentrated in the young adult bracket (18-24 years old) and is characterized by high general video game literacy, compounded by a gender imbalance (18 male and 9 female subjects). Furthermore, the sample’s very low familiarity with the specific text adventure genre – while proving valuable during testing to evaluate system accessibility – does not allow for determining a priori whether veteran interactive fiction players would respond similarly to the same uncertainty mechanics.

Second, the adopted experimental design is structured as a single-arm study, lacking a control group. Although this choice is consistent with the preliminary validation phases of a prototype, the absence of a comparative baseline – such as a version of the same text adventure without opacity and resource management mechanics – prevents quantitatively measuring the delta in effectiveness between the proposed model and conventional game design paradigms. The results demonstrate the internal consistency of the system and the presence of expected phenomena, but they do not allow for absolute comparative inferences.

A further critical issue lies in the adopted playtesting modality. The choice to conduct the play sessions in an unsupervised manner was primarily dictated by logistical and organizational constraints, linked to the geographical dispersion of the sample and the difficulty of setting up multiple sessions in a controlled environment. This practical necessity was coupled with the desire to avoid altering player behavior due to the awareness of being observed. However, this lack of direct control made it impossible to quantify the impact of any external confounding variables. Interruptions during gameplay, physiological drops in attention, or distractions stemming from the home environment might have introduced statistical “noise” into sensitive metrics such as interaction times, reading attention, and the subjective perception of cognitive fatigue.

Finally, the analysis highlighted how a specific user experience issue related to the camp mechanic acted as a confounding variable in the measurement of RQ4 (see Section 4.4). Although the rest mechanic was illustrated via an introductory text panel at the start of the experience, the absence of a contextual reminder at the exact moment the mechanic was first unlocked likely impaired information retention for a considerable portion of the sample (11 testers). This factor blurred the line between a deliberate strategic choice (rationing resources out of fear of the unknown) and simply forgetting the mechanic. This issue demonstrates how the empirical evaluation of systemic reasoning requires, in future studies, an interface designed to support player action at the exact moment it is enabled.

Chapter 5

Conclusions and Future Work

5.1 Conclusions

This thesis aimed to operationalize the theoretical CPS-GFC framework [3] by developing a serious game rooted in commercial video game mechanics. Building on the established premise that commercial titles effectively stimulate Complex Problem Solving skills – an area where traditional serious games often fail to engage users or support genuine gameplay-learning – this study sought to specifically validate the framework. By designing, implementing, and empirically validating a text-based narrative adventure built entirely on these guidelines and the strategic modulation of uncertainty via IMS, the research provides solid evidence supporting the feasibility of this hybrid approach.

Data analysis from playtesting sessions supported most initial hypotheses, revealing that mechanical friction does not obstruct the user but rather fosters learning. Notably, the triggering of prospective learning emerged with strong statistical significance: players exposed to adverse consequences from opaque interactions demonstrated a marked ability to readapt their mental models, transferring learned implicit rules to overcome novel challenges later in the simulation.

From a narrative and ethical perspective, inextricably intertwining mathematical resource management with character development generated high engagement. Qualitative and quantitative data highlighted how narrative immersion pushed users to take full responsibility for tactical choices, accepting irreversible consequences without perceiving the system as arbitrarily punitive.

Nevertheless, the study highlighted intrinsic difficulties in measuring complex constructs like long-term systemic reasoning. As evidenced by cross-referencing telemetry and open-ended responses, the boundary between strategic planning

(resource rationing) and interface vulnerabilities (forgetting camp access due to a lack of contextual feedback upon unlocking) remains blurred.

In summary, this thesis contributes an empirical baseline demonstrating how typical entertainment mechanics can qualitatively elevate CPS-dedicated serious games. Rather than perceiving the experience as a tedious “task”, participants were deeply drawn into the narrative ecosystem. The prototype indicates that designing serious games to train CPS skills does not require sacrificing fun for explicit didactics (e.g., intrusive tutorials). Instead, challenging users via the CPS-GFC framework in intrinsically motivating environments is a promising path for creating educational artifacts that are cognitively rigorous and deeply engaging.

5.2 Future Works

In light of the obtained results and methodological limitations discussed in Chapter 4, several directions for evolving and expanding this research emerge, both for empirical consolidation and design innovation:

- **User experience optimization:** The most immediate future development involves refining the digital artifact based on playtesting feedback. To rigorously isolate the user’s strategic intent, onboarding systems for silent mechanics must be updated. Implementing non-intrusive contextual feedback that activates only when an action (like setting up camp) becomes tangibly available would offload the player’s working memory, ensuring telemetric measurements exclusively reflect CPS choices rather than interface shortcomings.
- **Removal of exploratory constraints and reintroduction of game over:** Due to this study’s exploratory nature and the need to preserve data integrity across just 27 users, extreme consequences of systemic failure were mitigated. The absence of a premature game over prevented excessive telemetry fragmentation, replacing game-ending states with narrative and statistical penalties. However, with a larger sample, reintroducing a definitive failure state would push the user’s assumption of decision-making responsibility to the limit. Giving a lethal weight to resource depletion would naturally increase the sense of danger, amplifying emotional engagement and maximizing the risk-reward dynamic.
- **Multimodal integration and challenge diversification:** From a game design perspective, while a purely text-based interface proved functional, expanding the framework would benefit from integrating visual infrastructure. As discussed in Chapter 3, relying exclusively on text risks limiting variety in unfamiliar challenges: iterating learning on previously introduced mechanics via written variants can cause repetitiveness. Introducing graphical and

exploratory elements distributes information across multiple sensory channels, making prospective learning more impactful and enabling the designer to focus on purely ludic or visual-spatial deductions, freeing them from the strictly ethical-narrative dimension.

- **Evolution of experimental design and sampling:** To overcome the limitations of a single-group exploratory study, the natural next step is adopting a comparative experimental design. Introducing a control group – administered a prototype stripped of uncertainty management mechanics (or where their impact is minimized) – allows for rigorous A/B testing. This approach isolates the specific causal effect of the CPS-GFC framework on decision-making. To ensure greater external validity, administering the test to a demographically stratified sample is crucial. Comparing novices and veterans of interactive fiction could reveal how prior literacy in specific ludic grammars modulates the perceived cognitive load.
- **Integration of objective biometric metrics:** Currently, evaluating cognitive effort and ethical distress relies on telemetric data and self-report questionnaires – valuable tools, but physiologically exposed to subjective biases. An impactful methodological development would involve moving playtesting from an unsupervised remote setting to a controlled lab environment, integrating basic biometric instruments. Using sensors to detect stress-related physiological variations would allow real-time tracking of when a player experiences cognitive overload peaks or emotional dissonance during adverse moral choices, enabling a more robust triangulation of psychometric data.

Appendix A

Survey and Evaluation Tools

This appendix presents the complete wording of the questionnaires administered to the players during the playtest sessions, divided into Pre-Test and Post-Test. (Please note that the original questionnaires were administered in Italian; an English translation is provided below).

A.1 Pre-Test Questionnaire

Section A: Demographics and Video Gaming Profile

1. First Name, Last Name, Age, Gender
2. I tend to maintain high levels of attention during prolonged reading of on-screen texts.
3. How often do you play video games? [Never / Rarely / Monthly / Weekly / Daily]
4. How often do you play titles based primarily on text or narrative (e.g., visual novels, text adventures)? [1-7 Likert Scale]
5. I have played titles that require managing limited resources or making complex strategic decisions. [1-7 Likert Scale]

Section B: Perception of Complexity

6. When playing, I tend to consider the long-term consequences of my actions rather than focusing solely on the immediate goal. [1-7 Likert Scale]
7. I am interested in understanding the unwritten rules and cause-and-effect systems within video games. [1-7 Likert Scale]

8. I find it stimulating to make moral or ambiguous decisions within a video game. [1-7 Likert Scale]
9. In survival video games, my first reaction to an unknown character is usually one of strong mistrust. [1-7 Likert Scale]

Section C: Cognitive Attitudes and Playstyle

10. I prefer having time to plan my strategy before acting, rather than having to react instinctively. [1-7 Likert Scale]
11. I find it stimulating when a game makes the consequences of my actions irreversible. [1-7 Likert Scale]
12. I enjoy interpreting hidden clues in descriptions to anticipate danger. [1-7 Likert Scale]
13. I prefer interacting with characters who exhibit ambiguous and unpredictable behavior. [1-7 Likert Scale]
14. I tend to sacrifice useful resources to make a morally “good” choice or help a character. [1-7 Likert Scale]
15. In video games, I tend to explore dangerous areas or attempt high-risk actions if the potential reward (resources, information) is very high. [1-7 Likert Scale]
16. I like it when unforeseen events disrupt my plans, forcing me to improvise. [1-7 Likert Scale]
17. When faced with a new mechanic without a tutorial, I find it rewarding to deduce the rules autonomously. [1-7 Likert Scale]
18. I usually try to memorize details or behavioral patterns, expecting them to be useful in the future. [1-7 Likert Scale]

A.2 Post-Test Questionnaire

Section A: Engagement and Overall Learning

1. I found the story and atmosphere of the game engaging. [1-7 Likert Scale]
2. I found the game mechanics (e.g., choices, resource management) interesting and stimulating. [1-7 Likert Scale]

3. My choices within the story felt significant and not purely aesthetic. [1-7 Likert Scale]
4. I perceived a clear cause-and-effect relationship between the decisions made and the evolution of the story. [1-7 Likert Scale]
5. I learned to recognize certain recurring patterns in the game without the need for explicit explanations. [1-7 Likert Scale]

Section B: Specific Experience Evaluation

6. The textual descriptions of the environment helped me anticipate imminent dangers before they manifested. [1-7 Likert Scale]
7. When faced with an unforeseen event, I had to abandon my initial plan and quickly adapt my strategy. [1-7 Likert Scale]
8. The narrative feedback of the game (e.g., the consequences described after an action) was useful for understanding my mistakes and correcting my subsequent strategy. [1-7 Likert Scale]
9. I faced situations where the information available to me was incomplete or ambiguous, forcing me to proceed with caution or look for other clues. [1-7 Likert Scale]
10. Before making an important decision, I carefully analyzed all the fragmentary information available to me. [1-7 Likert Scale]
11. At branching choices, I evaluated whether the risk of a path was justified by the potential gain (strategic or moral). [1-7 Likert Scale]
12. I perceived that decisions made in the early stages had concrete and irreversible consequences in the advanced stages. [1-7 Likert Scale]
13. The information learned by interacting with characters proved useful for managing subsequent situations. [1-7 Likert Scale]
14. I had difficulty predicting the characters' reactions, which made the interactions more realistic and tense. [1-7 Likert Scale]
15. I happened to feel a conflict between the rational goal of conserving resources and an ethical desire (e.g., helping a character, avoiding violence). [1-7 Likert Scale]
16. The evolution of my relationship with characters considered "enemies" influenced my tactical choices. [1-7 Likert Scale]

17. I rationally exploited moments of calm to reorganize my resources in anticipation of future challenges. [1-7 Likert Scale]
18. When faced with new challenges whose exact rules I did not know, I was able to deduce a solution based on previous clues. [1-7 Likert Scale]
19. In managing my inventory, I tried to ration specific resources anticipating future scenarios, rather than consuming them to solve only immediate problems. [1-7 Likert Scale]
20. I decided to intentionally face high-risk situations to gain a tactical advantage. [1-7 Likert Scale]
21. Even without graphics, the textual descriptions allowed me to orient myself and clearly imagine the surrounding environment. [1-7 Likert Scale]
22. Despite the textual nature, I felt an emotional involvement in the events. [1-7 Likert Scale]
23. I felt that the outcomes of the story depended on my choices and not on chance. [1-7 Likert Scale]
24. The game stimulated active reflection on the humanity of the enemy. [1-7 Likert Scale]

Section C: Open-Ended Questions

25. Describe a specific moment in the game when you had to change your strategy due to an unforeseen event or new information. [Free text]
26. Did you ever make a strategically disadvantageous decision in order to act according to an ethical principle? If so, which one and why? [Free text]
27. Was there a moment when you used previously learned information to solve a new problem? Give an example. [Free text]

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